Solve Lambda Vacuum Field Equations

Case 6D_1E : 4D spacetime (t,r,θ,ϕ) + two individually curled axis (ψ_4,ψ_5)
1: Curled geometries: Circles
E: Curled metric - a[t]^2 $\mathcal{D}_4$[r] $p_4{}^2$ d$\psi_4{}^2$ – a[t]^2 $\mathcal{D}_4$[r] $p_5{}^2$ d$\psi_5{}^2$
3-space metric: -B(t,r) dr^2 – $\mathbb{C}$(t,r) r^2 dθ^2 – $\mathbb{C}$(t,r) r^2 sin$^2\theta$ dϕ^2
Time metric: A(t,r) dt^2

Line Element

Out[5908]=

$$ds^2 = A\ dt^2 - B\ dr^2 - r^2\,\mathbb{C}\ d\theta^2 - r^2\,\mathbb{C}\,\mathrm{Sin}[\theta]^2\ d\phi^2 - a^2\,p_4^2\,\mathcal{D}_4\ d\psi_4^2 - a^2\,p_5^2\,\mathcal{D}_4\ d\psi_5^2$$

Metric Tensor Matrices

Out[5914]=

$$g:\ (g_{\mu\nu}) = \begin{pmatrix} \frac{A}{c^2} & 0 & 0 & 0 & 0 & 0 \\ 0 & -B & 0 & 0 & 0 & 0 \\ 0 & 0 & -r^2\,\mathbb{C} & 0 & 0 & 0 \\ 0 & 0 & 0 & -r^2\,\mathbb{C}\,\mathrm{Sin}[\theta]^2 & 0 & 0 \\ 0 & 0 & 0 & 0 & -a^2\,p_4^2\,\mathcal{D}_4 & 0 \\ 0 & 0 & 0 & 0 & 0 & -a^2\,p_5^2\,\mathcal{D}_4 \end{pmatrix}$$

$$G:\ (g^{\mu\nu}) = \begin{pmatrix} \frac{c^2}{A} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{B} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{r^2\,\mathbb{C}} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\mathrm{Csc}[\theta]^2}{r^2\,\mathbb{C}} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{1}{a^2\,p_4^2\,\mathcal{D}_4} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{a^2\,p_5^2\,\mathcal{D}_4} \end{pmatrix}$$

Case 6D_1E: 6D = 4D spacetime (t,r,θ,ϕ) + two individually curled axis (ψ_4,ψ_5)
$g_{00}=\frac{A(t,r)}{c^2}$, $g_{11}=-B(t,r)$, $g_{22}=-\mathbb{C}(t,r)r^2$, $g_{33}=-\mathbb{C}(t,r)r^2\sin^2\theta$
$g_{44}=-a(t)^2\mathcal{D}_4(r)p_4{}^2$, $g_{55}=-a(t)^2\mathcal{D}_4(r)p_5{}^2$, a(t) & $\mathcal{D}_4$(r) not constant

The Lambda field equations can be represented as $\frac{R_{\mu\nu}}{g_{\mu\nu}} = \Lambda$:

$$R_{00}:\ \frac{2\,c^2\,\ddot{a}}{a\,A} - \frac{c^2\,\dot{a}\,\dot{A}}{a\,A^2} - \frac{c^2\,\dot{A}\,\dot{B}}{4\,A^2\,B} - \frac{c^2\,\dot{B}^2}{4\,A\,B^2} + \frac{c^2\,\ddot{B}}{2\,A\,B} - \frac{c^2\,\dot{A}\,\dot{\mathbb{C}}}{2\,A^2\,\mathbb{C}} - \frac{c^2\,\dot{\mathbb{C}}^2}{2\,A\,\mathbb{C}^2} + \frac{c^2\,\ddot{\mathbb{C}}}{A\,\mathbb{C}} - \frac{A'}{A\,B\,r} + \frac{(A')^2}{4\,A^2\,B} + \frac{A'\,B'}{4\,A\,B^2} - \frac{A'\,\mathbb{C}'}{2\,A\,B\,\mathbb{C}} - \frac{A'\,\mathcal{D}_4{}'}{2\,A\,B\,\mathcal{D}_4} - \frac{A''}{2\,A\,B} = \Lambda$$

$$R_{01}: \quad -\frac{\dot{B}}{B\,r}+\frac{\dot{\mathbb{C}}}{r\,\mathbb{C}}-\frac{\dot{a}\,A'}{a\,A}-\frac{\dot{\mathbb{C}}\,A'}{2\,A\,\mathbb{C}}-\frac{\dot{B}\,\mathbb{C}'}{2\,B\,\mathbb{C}}-\frac{\dot{\mathbb{C}}\,\mathbb{C}'}{2\,\mathbb{C}^2}+\frac{\dot{\mathbb{C}}'}{\mathbb{C}}+\frac{\dot{a}\,\mathbb{D}_4'}{a\,\mathbb{D}_4}-\frac{\dot{B}\,\mathbb{D}_4'}{2\,B\,\mathbb{D}_4} = 0$$

$$R_{11}: \quad \frac{c^2\,\dot{a}\,\dot{B}}{a\,A\,B}-\frac{c^2\,\dot{A}\,\dot{B}}{4\,A^2\,B}-\frac{c^2\,\dot{B}^2}{4\,A\,B^2}+\frac{c^2\,\ddot{B}}{2\,A\,B}+\frac{c^2\,\dot{B}\,\dot{\mathbb{C}}}{2\,A\,B\,\mathbb{C}}+\frac{(A')^2}{4\,A^2\,B}+\frac{B'}{B^2\,r}+$$
$$\frac{A'\,B'}{4\,A\,B^2}-\frac{2\,\mathbb{C}'}{B\,r\,\mathbb{C}}+\frac{B'\,\mathbb{C}'}{2\,B^2\,\mathbb{C}}+\frac{(\mathbb{C}')^2}{2\,B\,\mathbb{C}^2}+\frac{B'\,\mathbb{D}_4'}{2\,B^2\,\mathbb{D}_4}+\frac{(\mathbb{D}_4')^2}{2\,B\,\mathbb{D}_4^2}-\frac{A''}{2\,A\,B}-\frac{\mathbb{C}''}{B\,\mathbb{C}}-\frac{\mathbb{D}_4''}{B\,\mathbb{D}_4} = \Lambda$$

$$R_{22}: \quad -\frac{1}{B\,r^2}+\frac{1}{r^2\,\mathbb{C}}+\frac{c^2\,\dot{a}\,\dot{\mathbb{C}}}{a\,A\,\mathbb{C}}-\frac{c^2\,\dot{A}\,\dot{\mathbb{C}}}{4\,A^2\,\mathbb{C}}+\frac{c^2\,\dot{B}\,\dot{\mathbb{C}}}{4\,A\,B\,\mathbb{C}}+\frac{c^2\,\ddot{\mathbb{C}}}{2\,A\,\mathbb{C}}-$$
$$\frac{A'}{2\,A\,B\,r}+\frac{B'}{2\,B^2\,r}-\frac{2\,\mathbb{C}'}{B\,r\,\mathbb{C}}-\frac{A'\,\mathbb{C}'}{4\,A\,B\,\mathbb{C}}+\frac{B'\,\mathbb{C}'}{4\,B^2\,\mathbb{C}}-\frac{\mathbb{D}_4'}{B\,r\,\mathbb{D}_4}-\frac{\mathbb{C}'\,\mathbb{D}_4'}{2\,B\,\mathbb{C}\,\mathbb{D}_4}-\frac{\mathbb{C}''}{2\,B\,\mathbb{C}} = \Lambda$$

$$R_{44}: \quad \frac{c^2\,\dot{a}^2}{a^2\,A}+\frac{c^2\,\ddot{a}}{a\,A}-\frac{c^2\,\dot{a}\,\dot{A}}{2\,a\,A^2}+\frac{c^2\,\dot{a}\,\dot{B}}{2\,a\,A\,B}+\frac{c^2\,\dot{a}\,\dot{\mathbb{C}}}{a\,A\,\mathbb{C}}-\frac{\mathbb{D}_4'}{B\,r\,\mathbb{D}_4}-\frac{A'\,\mathbb{D}_4'}{4\,A\,B\,\mathbb{D}_4}+\frac{B'\,\mathbb{D}_4'}{4\,B^2\,\mathbb{D}_4}-\frac{\mathbb{C}'\,\mathbb{D}_4'}{2\,B\,\mathbb{C}\,\mathbb{D}_4}-\frac{\mathbb{D}_4''}{2\,B\,\mathbb{D}_4} = \Lambda$$

$$R_{55}: \quad \frac{c^2\,\dot{a}^2}{a^2\,A}+\frac{c^2\,\ddot{a}}{a\,A}-\frac{c^2\,\dot{a}\,\dot{A}}{2\,a\,A^2}+\frac{c^2\,\dot{a}\,\dot{B}}{2\,a\,A\,B}+\frac{c^2\,\dot{a}\,\dot{\mathbb{C}}}{a\,A\,\mathbb{C}}-\frac{\mathbb{D}_4'}{B\,r\,\mathbb{D}_4}-\frac{A'\,\mathbb{D}_4'}{4\,A\,B\,\mathbb{D}_4}+\frac{B'\,\mathbb{D}_4'}{4\,B^2\,\mathbb{D}_4}-\frac{\mathbb{C}'\,\mathbb{D}_4'}{2\,B\,\mathbb{C}\,\mathbb{D}_4}-\frac{\mathbb{D}_4''}{2\,B\,\mathbb{D}_4} = \Lambda$$

We observed in Notebook 6D_1 that R_{33} is redundant
and, so, R_{33} is omitted here and in all of the 6D notebooks.
We observe that R_{55}, also, is redundant, the same as R_{44}.

The remaining equations are very similar to Case 5D_1A. Thus,
we use the Case 5D approach as a guide to solving these equations.

a: $B(t,r) = a(t)^2 B(r)$, $\mathbb{C}(t,r) = a(t)^2 \mathbb{C}(r)$

The field equations become

$$R_{00}: \quad \frac{5\,c^2\,\ddot{a}}{a\,A}-\frac{5\,c^2\,\dot{a}\,\dot{A}}{2\,a\,A^2}-\frac{A'}{a^2\,A\,r\,B}+\frac{(A')^2}{4\,a^2\,A^2\,B}-\frac{A'\,\mathbb{C}'}{2\,a^2\,A\,\mathbb{C}\,B}+\frac{A'\,B'}{4\,a^2\,A\,B^2}-\frac{A'\,\mathbb{D}_4'}{2\,a^2\,A\,B\,\mathbb{D}_4}-\frac{A''}{2\,a^2\,A\,B} = \Lambda$$

$$R_{01}: \quad -\frac{2\,\dot{a}\,A'}{a\,A} = 0$$

$$R_{11}: \quad \frac{4\,c^2\,\dot{a}^2}{a^2\,A}+\frac{c^2\,\ddot{a}}{a\,A}-\frac{c^2\,\dot{a}\,\dot{A}}{2\,a\,A^2}+\frac{(A')^2}{4\,a^2\,A^2\,B}-\frac{2\,\mathbb{C}'}{a^2\,r\,\mathbb{C}\,B}+\frac{(\mathbb{C}')^2}{2\,a^2\,\mathbb{C}^2\,B}+\frac{B'}{a^2\,r\,B^2}+$$
$$\frac{A'\,B'}{4\,a^2\,A\,B^2}+\frac{\mathbb{C}'\,B'}{2\,a^2\,\mathbb{C}\,B^2}+\frac{B'\,\mathbb{D}_4'}{2\,a^2\,B^2\,\mathbb{D}_4}+\frac{(\mathbb{D}_4')^2}{2\,a^2\,B\,\mathbb{D}_4^2}-\frac{A''}{2\,a^2\,A\,B}-\frac{\mathbb{C}''}{a^2\,\mathbb{C}\,B}-\frac{\mathbb{D}_4''}{a^2\,B\,\mathbb{D}_4} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2\,r^2\,\mathbb{C}}-\frac{1}{a^2\,r^2\,B}+\frac{4\,c^2\,\dot{a}^2}{a^2\,A}+\frac{c^2\,\ddot{a}}{a\,A}-\frac{c^2\,\dot{a}\,\dot{A}}{2\,a\,A^2}-\frac{A'}{2\,a^2\,A\,r\,B}-$$
$$\frac{2\,\mathbb{C}'}{a^2\,r\,\mathbb{C}\,B}-\frac{A'\,\mathbb{C}'}{4\,a^2\,A\,\mathbb{C}\,B}+\frac{B'}{2\,a^2\,r\,B^2}+\frac{\mathbb{C}'\,B'}{4\,a^2\,\mathbb{C}\,B^2}-\frac{\mathbb{D}_4'}{a^2\,r\,B\,\mathbb{D}_4}-\frac{\mathbb{C}'\,\mathbb{D}_4'}{2\,a^2\,\mathbb{C}\,B\,\mathbb{D}_4}-\frac{\mathbb{C}''}{2\,a^2\,\mathbb{C}\,B} = \Lambda$$

$$R_{44}: \quad \frac{4\,c^2\,\dot{a}^2}{a^2\,A}+\frac{c^2\,\ddot{a}}{a\,A}-\frac{c^2\,\dot{a}\,\dot{A}}{2\,a\,A^2}-\frac{\mathbb{D}_4'}{a^2\,r\,B\,\mathbb{D}_4}-\frac{A'\,\mathbb{D}_4'}{4\,a^2\,A\,B\,\mathbb{D}_4}-\frac{\mathbb{C}'\,\mathbb{D}_4'}{2\,a^2\,\mathbb{C}\,B\,\mathbb{D}_4}+\frac{B'\,\mathbb{D}_4'}{4\,a^2\,B^2\,\mathbb{D}_4}-\frac{\mathbb{D}_4''}{2\,a^2\,B\,\mathbb{D}_4} = \Lambda$$

Since $a(t)$ is not allowed to be a constant in Case a, then $\dot{a} \neq 0$, and from R_{01} we get that A is a function solely of t:

$$A(t,r) = \mathcal{A}(t)$$

R_{01} is eliminated, and the remaining equations become

R_{00}:
$$\frac{5 c^2 \ddot{a}}{a \mathcal{A}} - \frac{5 c^2 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2} = \Lambda$$

R_{11}:
$$\frac{4 c^2 \dot{a}^2}{a^2 \mathcal{A}} + \frac{c^2 \ddot{a}}{a \mathcal{A}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2} - \frac{2 C'}{a^2 r C B} + \frac{(C')^2}{2 a^2 C^2 B} + \frac{B'}{a^2 r B^2} + \frac{C' B'}{2 a^2 C B^2} + \frac{B' D_4'}{2 a^2 B^2 D_4} + \frac{(D_4')^2}{2 a^2 B D_4^2} - \frac{C''}{a^2 C B} - \frac{D_4''}{a^2 B D_4} = \Lambda$$

R_{22}:
$$\frac{1}{a^2 r^2 C} - \frac{1}{a^2 r^2 B} + \frac{4 c^2 \dot{a}^2}{a^2 \mathcal{A}} + \frac{c^2 \ddot{a}}{a \mathcal{A}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2} - \frac{2 C'}{a^2 r C B} + \frac{B'}{2 a^2 r B^2} + \frac{C' B'}{4 a^2 C B^2} - \frac{D_4'}{a^2 r B D_4} - \frac{C' D_4'}{2 a^2 C B D_4} - \frac{C''}{2 a^2 C B} = \Lambda$$

R_{44}:
$$\frac{4 c^2 \dot{a}^2}{a^2 \mathcal{A}} + \frac{c^2 \ddot{a}}{a \mathcal{A}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2} - \frac{D_4'}{a^2 r B D_4} - \frac{C' D_4'}{2 a^2 C B D_4} + \frac{B' D_4'}{4 a^2 B^2 D_4} - \frac{D_4''}{2 a^2 B D_4} = \Lambda$$

We can solve R_{00} for $\mathcal{A}(t)$:

$$\mathcal{A}(t) = \frac{5 c^2 \dot{a}^2}{5 c^2 k + a^2 \Lambda}$$ where k is a constant of integration.

Plugging this gives us 3 equations in the 3 unknowns $B(r)$, $C(r)$, and $D_4(r)$:

R_{11}:
$$\frac{4 c^2 k}{a^2} + \Lambda - \frac{2 C'}{a^2 r C B} + \frac{(C')^2}{2 a^2 C^2 B} + \frac{B'}{a^2 r B^2} + \frac{C' B'}{2 a^2 C B^2} + \frac{B' D_4'}{2 a^2 B^2 D_4} + \frac{(D_4')^2}{2 a^2 B D_4^2} - \frac{C''}{a^2 C B} - \frac{D_4''}{a^2 B D_4} = \Lambda$$

R_{22}:
$$\frac{4 c^2 k}{a^2} + \frac{1}{a^2 r^2 C} - \frac{1}{a^2 r^2 B} + \Lambda - \frac{2 C'}{a^2 r C B} + \frac{B'}{2 a^2 r B^2} + \frac{C' B'}{4 a^2 C B^2} - \frac{D_4'}{a^2 r B D_4} - \frac{C' D_4'}{2 a^2 C B D_4} - \frac{C''}{2 a^2 C B} = \Lambda$$

R_{44}:
$$\frac{4 c^2 k}{a^2} + \Lambda - \frac{D_4'}{a^2 r B D_4} - \frac{C' D_4'}{2 a^2 C B D_4} + \frac{B' D_4'}{4 a^2 B^2 D_4} - \frac{D_4''}{2 a^2 B D_4} = \Lambda$$

These equations, individually, are too difficult to be solved. However, $R_{11}-2R_{22}-2R_{44}$ can be algebraically solved for B or Dsolved for either of C or D_4. We choose to solve for B:

$R_{11}-2R_{22}-2R_{44}-$:
$$-\frac{12 c^2 k}{a^2} - \frac{2}{a^2 r^2 C} + \frac{2}{a^2 r^2 B} - 3 \Lambda + \frac{2 C'}{a^2 r C B} + \frac{(C')^2}{2 a^2 C^2 B} + \frac{4 D_4'}{a^2 r B D_4} + \frac{2 C' D_4'}{a^2 C B D_4} + \frac{(D_4')^2}{2 a^2 B D_4^2} = -3 \Lambda$$

$$B(r) = \frac{4 C^2 D_4^2 + 4 r C D_4^2 C' + r^2 D_4^2 (C')^2 + 8 r C^2 D_4 D_4' + 4 r^2 C D_4 C' D_4' + r^2 C^2 (D_4')^2}{4 C \left(1 + 6 c^2 k r^2 C\right) D_4^2}$$

Plugging back in, the equations are too lengthy to include here.
However, $R_{11}-R_{44} = 0$, so the LHS can be put over a common denominator and then the denominator can be discarded. The resulting LHS has a factor of $(2\mathbb{C}+r\mathbb{C}')$, so we can easily find at least one solution:

$R_{11}-R_{44}$: lhs14 = rhs14

Setting $2\mathbb{C}+r\mathbb{C}'=0$ and solving, we get:

$$\mathbb{C}(r) = \frac{K}{r^2} \quad \text{where } K \text{ is a constant of integration}$$

$$\Rightarrow \quad \mathbb{B}(r) = \frac{K\,(\mathbb{D}_4')^2}{4\left(1+6\,c^2\,k\,K\right)\mathbb{D}_4^2}$$

When we plug $\mathbb{B}(r)$ and $\mathbb{C}(r)$ back in, the equations become identical:

$$R_{11}: \quad -\frac{8\,c^2\,k}{a^2}-\frac{2}{a^2\,K}+\Lambda = \Lambda, \quad \text{or} \quad -\frac{8\,c^2\,k}{a^2}-\frac{2}{a^2\,K} = 0$$

$$R_{22}: \quad \frac{4\,c^2\,k}{a^2}+\frac{1}{a^2\,K}+\Lambda = \Lambda, \quad \text{or} \quad \frac{4\,c^2\,k}{a^2}+\frac{1}{a^2\,K} = 0$$

$$R_{44}: \quad -\frac{8\,c^2\,k}{a^2}-\frac{2}{a^2\,K}+\Lambda = \Lambda, \quad \text{or} \quad -\frac{8\,c^2\,k}{a^2}-\frac{2}{a^2\,K} = 0$$

- R_{44} is the same as R_{11}
- $R_{11} = -2\ R_{22}$

Solving R_{22} for k yields

$$k = -\frac{1}{4\,c^2\,K}$$

$$\Rightarrow \quad \mathcal{A}(t) = \frac{20\,c^2\,K\,\dot{a}^2}{-5+4\,a^2\,K\,\Lambda}$$

$$\Rightarrow \quad \mathbb{B}(r) = -\frac{K\,(\mathbb{D}_4')^2}{2\,\mathbb{D}_4^2}$$

We have found an exact solution to the Λ field equations:

$$A(t,r) = \mathcal{A}(t) = \frac{20\,c^2\,K\,\dot{a}[t]^2}{-5+4\,K\,\Lambda\,a[t]^2} \quad \text{where } K \text{ is a free constant,}$$

$$\mathbb{B}(t,r) = a(t)^2\mathbb{B}(r) = a(t)^2-\frac{K\,\mathbb{D}_4'[r]^2}{2\,\mathbb{D}_4[r]^2},$$

$$\mathbb{C}(t,r) = a(t)^2\mathbb{C}(r) = a(t)^2\frac{K}{r^2},$$

$$\mathbb{D}_4(t,r) = \mathbb{D}_5(t,r) = a(t)^2\mathbb{D}_4(r) \quad \text{where } \mathbb{D}_4(r) \text{ is a free parameter,}$$

and line element

$$ds^2 = \frac{20\,c^2\,K\,a'[t]^2}{-5+4\,K\,\Lambda\,a[t]^2}\,dt^2 + \frac{K\,a[t]^2\,D_4'[r]^2}{2\,D_4[r]^2}\,dr^2 - K\,a[t]^2\,d\theta^2 - K\,a[t]^2\,\mathrm{Sin}[\theta]^2\,d\phi^2 - a[t]^2\,p_4^2\,D_4[r]\,d\psi_4^2 - a[t]^2\,p_5^2\,D_4[r]\,d\psi_5^2$$

a-II Solve Case a for a(t) when A(t,r) = c^2

Solving $\frac{20\,c^2\,K\,\dot{a}[t]^2}{-5+4\,K\,\Lambda\,a[t]^2} = c^2$ yields

$$a(t) = -\frac{\sqrt{5}\,\mathrm{Sin}\left[\frac{\sqrt{\Lambda}\,(\upsilon \pm i\,t)}{\sqrt{5}}\right]}{2\,\sqrt{K}\,\sqrt{\Lambda}}$$ where υ is a free constant.

The normal next steps are to explore 3 subcases of Case a:

(a_II) A(t,r)=c^2

(ai) B(r)=1 [i.e., B(t,r)=a(t)^2]

(aii) C(r)=1 [i.e., $\mathbb{C}$(t,r)=a(t)^2]

a_II: B(t,r)=a(t)^{2}B(r), $\mathbb{C}$(t,r)=a(t)^{2}C(r), A(t,r)=c^2

Even though this case is a subset of the solution to Case a, it is possible we may find a different solution by setting A(t,r)=c^2 prior to solving the ODEs. Re-doing Case a from scratch, the field equations become

$$R_{00}:\quad \frac{5\,\ddot{a}}{a} = \Lambda$$

$$R_{11}:\quad \frac{4\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} - \frac{2\,C'}{a^2\,r\,C\,B} + \frac{(C')^2}{2\,a^2\,C^2\,B} + \frac{B'}{a^2\,r\,B^2} + \frac{C'\,B'}{2\,a^2\,C\,B^2} + \frac{B'\,D_4'}{2\,a^2\,B^2\,D_4} + \frac{(D_4')^2}{2\,a^2\,B\,D_4^2} - \frac{C''}{a^2\,C\,B} - \frac{D_4''}{a^2\,B\,D_4} = \Lambda$$

$$R_{22}:\quad \frac{1}{a^2\,r^2\,C} - \frac{1}{a^2\,r^2\,B} + \frac{4\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} - \frac{2\,C'}{a^2\,r\,C\,B} + \frac{B'}{2\,a^2\,r\,B^2} + \frac{C'\,B'}{4\,a^2\,C\,B^2} - \frac{D_4'}{a^2\,r\,B\,D_4} - \frac{C'\,D_4'}{2\,a^2\,C\,B\,D_4} - \frac{C''}{2\,a^2\,C\,B} = \Lambda$$

$$R_{44}:\quad \frac{4\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} - \frac{D_4'}{a^2\,r\,B\,D_4} - \frac{C'\,D_4'}{2\,a^2\,C\,B\,D_4} + \frac{B'\,D_4'}{4\,a^2\,B^2\,D_4} - \frac{D_4''}{2\,a^2\,B\,D_4} = \Lambda$$

Solving R_{00} yields

$$a(t) = e^{\frac{t\sqrt{\Lambda}}{\sqrt{5}}}\,\alpha + e^{-\frac{t\sqrt{\Lambda}}{\sqrt{5}}}\,\beta$$ where α and β are constants of integration.

This a(t) is a generalization of the a(t) we found in Case a-II, unless either α or β is zero.

We can also algebraically solve $R_{11}-2R_{22}-2R_{44}$ for B(r):

$R_{11}-2R_{22}-2R_{44}$:

$$-\frac{2}{a^2 r^2 C}+\frac{2}{a^2 r^2 B}-\frac{12\dot{a}^2}{a^2}-\frac{3\ddot{a}}{a}+\frac{2C'}{a^2 r C B}+\frac{(C')^2}{2a^2 C^2 B}+\frac{4D_4'}{a^2 r B D_4}+\frac{2C'D_4'}{a^2 C B D_4}+\frac{(D_4')^2}{2a^2 B D_4^2}=-3\Lambda$$

$$B(r)=\frac{-4C^2D_4^2-4rCD_4^2C'-r^2D_4^2(C')^2-8rC^2D_4D_4'-4r^2CD_4C'D_4'-r^2C^2(D_4')^2}{2C\left(-2+3a^2r^2C\Lambda-12r^2C\dot{a}^2-3ar^2C\ddot{a}\right)D_4^2}$$

In order for $B(r)$ to strictly be a function of r, the t terms in the denominator must disappear. There are no terms in the numerator to cancel them out, so the denominator terms must be self-canceling. It turns out they are. Call the denominator expression in the parentheses $p(t,r)$:

$$p(t,r)=-2+3a^2r^2C\Lambda-12r^2C\dot{a}^2-3ar^2C\ddot{a}$$

When we plug in the formula above for $a(t)$, this simplifies to

$$p(t,r)=-2+\frac{48}{5}r^2\alpha\beta\Lambda C[r]$$

$$\Rightarrow\quad B(r)=-\frac{5\left(r^2D_4^2(C')^2+4rCD_4C'(D_4+rD_4')+C^2\left(4D_4^2+8rD_4D_4'+r^2(D_4')^2\right)\right)}{4C\left(-5+24r^2C\alpha\beta\Lambda\right)D_4^2},$$

strictly a function of r.

We are imitating the Case 5D_1A methodology. When we plug $B(r)$ into the remaining equations, they are each even longer than in Case 5D_1A, too long to display here. As in Case 5D_1A, we set $\beta=0$ to shorten the equations, and we put the left-hand sides of R_{11}, R_{22}, and R_{44} over their common denominators, which turn out to be the same common denominator for each of three equations.

The three equations are still to long to display here. $R_{11}-R_{22}$ is by far the simplest of the linear combination of the equations. We can ignore the LHS denominator of $R_{11}-R_{22}$ because the RHS is zero. The resulting numerator is more complex than in Case5D_1A but short enough to display here:

$R_{11}-R_{22}$: $-24rC^3D_4^2(D_4')^2-12r^2C^2D_4^2C'(D_4')^2-9r^3CD_4^2(C')^2(D_4')^2-12r^2C^3D_4(D_4')^3-6r^3C^2D_4C'(D_4')^3-3r^3C^3(D_4')^4+6r^3C^2D_4^2(D_4')^2C''-12r^2C^3D_4^2D_4'D_4''-6r^3C^2D_4^2C'D_4'D_4''$
$=0$

However, unlike in Case 5D_1A, I am unable to solve this equation for either C or D_4.

I am unable to solve this case.

ai: $B(t,r)=a[t]^2$, $\mathbb{C}(t,r)=a(t)^2 C(r)$

Starting fresh, the field equations become

$$R_{00}: \quad \frac{5c^2\ddot{a}}{aA} - \frac{5c^2\dot{a}\dot{A}}{2aA^2} - \frac{A'}{a^2 A r} + \frac{(A')^2}{4a^2A^2} - \frac{A'C'}{2a^2AC} - \frac{A'D_4'}{2a^2AD_4} - \frac{A''}{2a^2A} = \Lambda$$

$$R_{01}: \quad -\frac{2\dot{a}A'}{aA} = 0$$

$$R_{11}: \quad \frac{4c^2\dot{a}^2}{a^2A} + \frac{c^2\ddot{a}}{aA} - \frac{c^2\dot{a}\dot{A}}{2aA^2} + \frac{(A')^2}{4a^2A^2} - \frac{2C'}{a^2rC} + \frac{(C')^2}{2a^2C^2} + \frac{(D_4')^2}{2a^2D_4^2} - \frac{A''}{2a^2A} - \frac{C''}{a^2C} - \frac{D_4''}{a^2D_4} = \Lambda$$

R_{22}:

$$-\frac{1}{a^2r^2} + \frac{1}{a^2r^2C} + \frac{4c^2\dot{a}^2}{a^2A} + \frac{c^2\ddot{a}}{aA} - \frac{c^2\dot{a}\dot{A}}{2aA^2} - \frac{A'}{2a^2Ar} - \frac{2C'}{a^2rC} - \frac{A'C'}{4a^2AC} - \frac{D_4'}{a^2rD_4} - \frac{C'D_4'}{2a^2CD_4} - \frac{C''}{2a^2C} = \Lambda$$

$$R_{44}: \quad \frac{4c^2\dot{a}^2}{a^2A} + \frac{c^2\ddot{a}}{aA} - \frac{c^2\dot{a}\dot{A}}{2aA^2} - \frac{D_4'}{a^2rD_4} - \frac{A'D_4'}{4a^2AD_4} - \frac{C'D_4'}{2a^2CD_4} - \frac{D_4''}{2a^2D_4} = \Lambda$$

We again get that

$A(t,r) = \mathcal{A}(t)$, a function strictly of t.

R_{01} is eliminated, and the remaining equations become

$$R_{00}: \quad \frac{5c^2\ddot{a}}{a\mathcal{A}} - \frac{5c^2\dot{a}\dot{\mathcal{A}}}{2a\mathcal{A}^2} = \Lambda$$

$$R_{11}: \quad \frac{4c^2\dot{a}^2}{a^2\mathcal{A}} + \frac{c^2\ddot{a}}{a\mathcal{A}} - \frac{c^2\dot{a}\dot{\mathcal{A}}}{2a\mathcal{A}^2} - \frac{2C'}{a^2rC} + \frac{(C')^2}{2a^2C^2} + \frac{(D_4')^2}{2a^2D_4^2} - \frac{C''}{a^2C} - \frac{D_4''}{a^2D_4} = \Lambda$$

$$R_{22}: \quad -\frac{1}{a^2r^2} + \frac{1}{a^2r^2C} + \frac{4c^2\dot{a}^2}{a^2\mathcal{A}} + \frac{c^2\ddot{a}}{a\mathcal{A}} - \frac{c^2\dot{a}\dot{\mathcal{A}}}{2a\mathcal{A}^2} - \frac{2C'}{a^2rC} - \frac{D_4'}{a^2rD_4} - \frac{C'D_4'}{2a^2CD_4} - \frac{C''}{2a^2C} = \Lambda$$

$$R_{44}: \quad \frac{4c^2\dot{a}^2}{a^2\mathcal{A}} + \frac{c^2\ddot{a}}{a\mathcal{A}} - \frac{c^2\dot{a}\dot{\mathcal{A}}}{2a\mathcal{A}^2} - \frac{D_4'}{a^2rD_4} - \frac{C'D_4'}{2a^2CD_4} - \frac{D_4''}{2a^2D_4} = \Lambda$$

We solve R_{00} for $\mathcal{A}(t)$ and again get:

$$\mathcal{A}(t) = \frac{5c^2\dot{a}^2}{5c^2k + a^2\Lambda}$$ where k is a constant of integration.

Plugging this in gives us 3 equations in the 2 unknowns $C(r)$ and $D_4(r)$:

$$R_{11}: \quad \frac{4c^2k}{a^2} + \Lambda - \frac{2C'}{a^2rC} + \frac{(C')^2}{2a^2C^2} + \frac{(D_4')^2}{2a^2D_4^2} - \frac{C''}{a^2C} - \frac{D_4''}{a^2D_4} = \Lambda$$

$$R_{22}: \quad \frac{4c^2k}{a^2} - \frac{1}{a^2r^2} + \frac{1}{a^2r^2C} + \Lambda - \frac{2C'}{a^2rC} - \frac{D_4'}{a^2rD_4} - \frac{C'D_4'}{2a^2CD_4} - \frac{C''}{2a^2C} = \Lambda$$

$$R_{44}: \quad \frac{4c^2k}{a^2} + \Lambda - \frac{D_4'}{a^2rD_4} - \frac{C'D_4'}{2a^2CD_4} - \frac{D_4''}{2a^2D_4} = \Lambda$$

We can combine the three equations to eliminate the second derivatves:

$$R_{11}-2R_{22}-2R_{44}: \quad -\frac{12\,c^2\,k}{a^2}+\frac{2}{a^2\,r^2}-\frac{2}{a^2\,r^2\,C}-3\,\Lambda+\frac{2\,C'}{a^2\,r\,C}+\frac{(C')^2}{2\,a^2\,C^2}+\frac{4\,D_4'}{a^2\,r\,D_4}+\frac{2\,C'\,D_4'}{a^2\,C\,D_4}+\frac{(D_4')^2}{2\,a^2\,D_4^2} = -3\,\Lambda$$

We are unable to solve this equation for C, but there is a solution for D_4:

$$D_4(r) = e^{\int_1^r \frac{-4\,C-2\,\rho\,C'+\sqrt{4\,C+12\,C^2+24\,c^2\,k\,C^2\,\rho^2+12\,C\,\rho\,C'+3\,\rho^2\,(C')^2}}{C\,\rho}\,d\rho}\,\gamma$$ where γ is a constant of integration.

When we plug this back into the equations, the integral conveniently disappears but, unfortunately, the equatons are too unwieldy for me to solve.

aii: $B(t,r)=a(t)^2 B(r)$, $C(t,r)=a[t]^2$

Starting fresh, the resulting field equations are :

$$R_{00}: \quad \frac{5\,c^2\,\ddot{a}}{a\,A}-\frac{5\,c^2\,\dot{a}\,\dot{A}}{2\,a\,A^2}-\frac{A'}{a^2\,A\,r\,B}+\frac{(A')^2}{4\,a^2\,A^2\,B}+\frac{A'\,B'}{4\,a^2\,A\,B^2}-\frac{A'\,D_4'}{2\,a^2\,A\,B\,D_4}-\frac{A''}{2\,a^2\,A\,B} = \Lambda$$

$$R_{01}: \quad -\frac{2\,\dot{a}\,A'}{a\,A} = 0$$

R_{11}:

$$\frac{4\,c^2\,\dot{a}^2}{a^2\,A}+\frac{c^2\,\ddot{a}}{a\,A}-\frac{c^2\,\dot{a}\,\dot{A}}{2\,a\,A^2}+\frac{(A')^2}{4\,a^2\,A^2\,B}+\frac{B'}{a^2\,r\,B^2}+\frac{A'\,B'}{4\,a^2\,A\,B^2}+\frac{B'\,D_4'}{2\,a^2\,B^2\,D_4}+\frac{(D_4')^2}{2\,a^2\,B\,D_4^2}-\frac{A''}{2\,a^2\,A\,B}-\frac{D_4''}{a^2\,B\,D_4} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2\,r^2}-\frac{1}{a^2\,r^2\,B}+\frac{4\,c^2\,\dot{a}^2}{a^2\,A}+\frac{c^2\,\ddot{a}}{a\,A}-\frac{c^2\,\dot{a}\,\dot{A}}{2\,a\,A^2}-\frac{A'}{2\,a^2\,A\,r\,B}+\frac{B'}{2\,a^2\,r\,B^2}-\frac{D_4'}{a^2\,r\,B\,D_4} = \Lambda$$

$$R_{44}: \quad \frac{4\,c^2\,\dot{a}^2}{a^2\,A}+\frac{c^2\,\ddot{a}}{a\,A}-\frac{c^2\,\dot{a}\,\dot{A}}{2\,a\,A^2}-\frac{D_4'}{a^2\,r\,B\,D_4}-\frac{A'\,D_4'}{4\,a^2\,A\,B\,D_4}+\frac{B'\,D_4'}{4\,a^2\,B^2\,D_4}-\frac{D_4''}{2\,a^2\,B\,D_4} = \Lambda$$

From R_{01} we again get that

$A(t,r) = \mathcal{A}[t]$, a function strictly of t.

R_{01} is eliminated, and the remaining equations become

$$R_{00}: \quad \frac{5\,c^2\,\ddot{a}}{a\,\mathcal{A}}-\frac{5\,c^2\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2} = \Lambda$$

$$R_{11}: \quad \frac{4\,c^2\,\dot{a}^2}{a^2\,\mathcal{A}}+\frac{c^2\,\ddot{a}}{a\,\mathcal{A}}-\frac{c^2\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2}+\frac{B'}{a^2\,r\,B^2}+\frac{B'\,D_4'}{2\,a^2\,B^2\,D_4}+\frac{(D_4')^2}{2\,a^2\,B\,D_4^2}-\frac{D_4''}{a^2\,B\,D_4} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2\,r^2}-\frac{1}{a^2\,r^2\,B}+\frac{4\,c^2\,\dot{a}^2}{a^2\,\mathcal{A}}+\frac{c^2\,\ddot{a}}{a\,\mathcal{A}}-\frac{c^2\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2}+\frac{B'}{2\,a^2\,r\,B^2}-\frac{D_4'}{a^2\,r\,B\,D_4} = \Lambda$$

$$R_{44}: \quad \frac{4\,c^2\,\dot{a}^2}{a^2\,\mathcal{A}}+\frac{c^2\,\ddot{a}}{a\,\mathcal{A}}-\frac{c^2\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2}-\frac{D_4'}{a^2\,r\,B\,D_4}+\frac{B'\,D_4'}{4\,a^2\,B^2\,D_4}-\frac{D_4''}{2\,a^2\,B\,D_4} = \Lambda$$

We solve R_{00} for $\mathcal{A}(t)$ and again get:

$$\mathcal{A}(t) = \frac{5\,c^2\,\dot{a}^2}{5\,c^2\,k + a^2\,\Lambda}$$ where k is a constant of integration.

Plugging this in gives us 3 equations in the 2 unknowns $B(r)$ and $D_4(r)$:

$$R_{11}: \quad \frac{4\,c^2\,k}{a^2} + \Lambda + \frac{B'}{a^2\,r\,B^2} + \frac{B'\,D_4'}{2\,a^2\,B^2\,D_4} + \frac{(D_4')^2}{2\,a^2\,B\,D_4^2} - \frac{D_4''}{a^2\,B\,D_4} = \Lambda$$

$$R_{22}: \quad \frac{4\,c^2\,k}{a^2} + \frac{1}{a^2\,r^2} - \frac{1}{a^2\,r^2\,B} + \Lambda + \frac{B'}{2\,a^2\,r\,B^2} - \frac{D_4'}{a^2\,r\,B\,D_4} = \Lambda$$

$$R_{44}: \quad \frac{4\,c^2\,k}{a^2} + \Lambda - \frac{D_4'}{a^2\,r\,B\,D_4} + \frac{B'\,D_4'}{4\,a^2\,B^2\,D_4} - \frac{D_4''}{2\,a^2\,B\,D_4} = \Lambda$$

At this point, there are many different approaches that can be tried. I pursued many but they all led to solutions for either B or D_4 that involved integrals, and none led subsequently to a solution for the other unknown.

These equations are too complex for me to solve.

b: B(t,r)=B(r), C(t,r)=C(r)

The field equations become:

$$R_{00}: \quad \frac{2\,c^2\,\ddot{a}}{a\,A} - \frac{c^2\,\dot{a}\,\dot{A}}{a\,A^2} - \frac{A'}{A\,r\,B} + \frac{(A')^2}{4\,A^2\,B} - \frac{A'\,C'}{2\,A\,C\,B} + \frac{A'\,B'}{4\,A\,B^2} - \frac{A'\,D_4'}{2\,A\,B\,D_4} - \frac{A''}{2\,A\,B} = \Lambda$$

$$R_{01}: \quad -\frac{\dot{a}\,A'}{a\,A} + \frac{\dot{a}\,D_4'}{a\,D_4} = 0$$

$$R_{11}: \quad \frac{(A')^2}{4\,A^2\,B} - \frac{2\,C'}{r\,C\,B} + \frac{(C')^2}{2\,C^2\,B} + \frac{B'}{r\,B^2} + \frac{A'\,B'}{4\,A\,B^2} + \frac{C'\,B'}{2\,C\,B^2} + \frac{B'\,D_4'}{2\,B^2\,D_4} + \frac{(D_4')^2}{2\,B\,D_4^2} - \frac{A''}{2\,A\,B} - \frac{C''}{C\,B} - \frac{D_4''}{B\,D_4} = \Lambda$$

$$R_{22}: \quad \frac{1}{r^2\,C} - \frac{1}{r^2\,B} - \frac{A'}{2\,A\,r\,B} - \frac{2\,C'}{r\,C\,B} - \frac{A'\,C'}{4\,A\,C\,B} + \frac{B'}{2\,r\,B^2} + \frac{C'\,B'}{4\,C\,B^2} - \frac{D_4'}{r\,B\,D_4} - \frac{C'\,D_4'}{2\,C\,B\,D_4} - \frac{C''}{2\,C\,B} = \Lambda$$

$$R_{44}: \quad \frac{c^2\,\dot{a}^2}{a^2\,A} + \frac{c^2\,\ddot{a}}{a\,A} - \frac{c^2\,\dot{a}\,\dot{A}}{2\,a\,A^2} - \frac{D_4'}{r\,B\,D_4} - \frac{A'\,D_4'}{4\,A\,B\,D_4} - \frac{C'\,D_4'}{2\,C\,B\,D_4} + \frac{B'\,D_4'}{4\,B^2\,D_4} - \frac{D_4''}{2\,B\,D_4} = \Lambda$$

Because $a(t) \neq$ constant, $\dot{a} \neq 0$. This allows us to simplify R_{01} by factoring out $\frac{\dot{a}}{a}$, and then we can solve for A:

$$R_{01}: \quad \frac{A'}{A} - \frac{D_4'}{D_4} = 0$$

$A(t,r) = \mathcal{A}[t]\,D_4[r]$ where $\mathcal{A}(t)$ is a new unknown.

The equations now become

$$R_{00}: \quad \frac{2\,c^2\,\ddot{a}}{a\,\mathcal{A}\,D_4} - \frac{c^2\,\dot{a}\,\dot{\mathcal{A}}}{a\,\mathcal{A}^2\,D_4} - \frac{D_4'}{r\,B\,D_4} - \frac{C'\,D_4'}{2\,C\,B\,D_4} + \frac{B'\,D_4'}{4\,B^2\,D_4} - \frac{(D_4')^2}{4\,B\,D_4^2} - \frac{D_4''}{2\,B\,D_4} = \Lambda$$

$$R_{11}: \quad -\frac{2\,C'}{r\,C\,B}+\frac{(C')^2}{2\,C^2\,B}+\frac{B'}{r\,B^2}+\frac{C'\,B'}{2\,C\,B^2}+\frac{3\,B'\,D_4'}{4\,B^2\,D_4}+\frac{3\,(D_4')^2}{4\,B\,D_4^2}-\frac{C''}{C\,B}-\frac{3\,D_4''}{2\,B\,D_4}=\Lambda$$

$$R_{22}: \quad \frac{1}{r^2\,C}-\frac{1}{r^2\,B}-\frac{2\,C'}{r\,C\,B}+\frac{B'}{2\,r\,B^2}+\frac{C'\,B'}{4\,C\,B^2}-\frac{3\,D_4'}{2\,r\,B\,D_4}-\frac{3\,C'\,D_4'}{4\,C\,B\,D_4}-\frac{C''}{2\,C\,B}=\Lambda$$

$$R_{44}: \quad \frac{c^2\,\dot{a}^2}{a^2\,\mathcal{A}\,D_4}+\frac{c^2\,\ddot{a}}{a\,\mathcal{A}\,D_4}-\frac{c^2\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2\,D_4}-\frac{D_4'}{r\,B\,D_4}-\frac{C'\,D_4'}{2\,C\,B\,D_4}+\frac{B'\,D_4'}{4\,B^2\,D_4}-\frac{(D_4')^2}{4\,B\,D_4^2}-\frac{D_4''}{2\,B\,D_4}=\Lambda$$

From R_{00}-R_{44} we can solve for $\mathcal{A}(t)$:

$$R_{00}\text{-}R_{44}: \quad -\frac{c^2\,\dot{a}^2}{a^2\,\mathcal{A}\,D_4}+\frac{c^2\,\ddot{a}}{a\,\mathcal{A}\,D_4}-\frac{c^2\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2\,D_4}=0$$

or

$$-\frac{\dot{a}^2}{a}+\ddot{a}-\frac{\dot{a}\,\dot{\mathcal{A}}}{2\,\mathcal{A}}=0$$

$$\mathcal{A}[t]=\frac{k\,\dot{a}[t]^2}{a[t]^2}$$ where k is a constant of integration.

Plugging this in, R_{44} becomes redundant, the same as R_{00}:

$$R_{00}: \quad \frac{2\,c^2}{k\,D_4}-\frac{D_4'}{r\,B\,D_4}-\frac{C'\,D_4'}{2\,C\,B\,D_4}+\frac{B'\,D_4'}{4\,B^2\,D_4}-\frac{(D_4')^2}{4\,B\,D_4^2}-\frac{D_4''}{2\,B\,D_4}=\Lambda$$

$$R_{11}: \quad -\frac{2\,C'}{r\,C\,B}+\frac{(C')^2}{2\,C^2\,B}+\frac{B'}{r\,B^2}+\frac{C'\,B'}{2\,C\,B^2}+\frac{3\,B'\,D_4'}{4\,B^2\,D_4}+\frac{3\,(D_4')^2}{4\,B\,D_4^2}-\frac{C''}{C\,B}-\frac{3\,D_4''}{2\,B\,D_4}=\Lambda$$

$$R_{22}: \quad \frac{1}{r^2\,C}-\frac{1}{r^2\,B}-\frac{2\,C'}{r\,C\,B}+\frac{B'}{2\,r\,B^2}+\frac{C'\,B'}{4\,C\,B^2}-\frac{3\,D_4'}{2\,r\,B\,D_4}-\frac{3\,C'\,D_4'}{4\,C\,B\,D_4}-\frac{C''}{2\,C\,B}=\Lambda$$

$$R_{44}: \quad \frac{2\,c^2}{k\,D_4}-\frac{D_4'}{r\,B\,D_4}-\frac{C'\,D_4'}{2\,C\,B\,D_4}+\frac{B'\,D_4'}{4\,B^2\,D_4}-\frac{(D_4')^2}{4\,B\,D_4^2}-\frac{D_4''}{2\,B\,D_4}=\Lambda$$

I tried to solve the individual equations as well as several algebraic combinations, with no success. The furthest I got was soving R_{00} for C:

$$C(r)=e^{\int_{1}^{r}\frac{1}{2}\left(-\frac{4}{\rho}+\frac{B'}{B}-\frac{4\,B\,(-2\,c^2+k\,\Lambda\,D_4)}{k\,D_4'}-\frac{D_4'}{D_4}-\frac{2\,D_4''}{D_4'}\right)d\rho}\,c_1$$

Plugging this back in, both R_{00} are R_{44} are satisfied, leaving just two equations in two unknowns:

$$R_{00}: \quad \Lambda=\Lambda$$

$$R_{11}: \quad \Lambda+\frac{2\,c^2}{k\,D_4}+\frac{5\,(B')^2}{8\,B^3}-\frac{8\,c^4\,B}{k^2\,(D_4')^2}+\frac{8\,c^2\,B\,\Lambda\,D_4}{k\,(D_4')^2}-\frac{2\,B\,\Lambda^2\,D_4^2}{(D_4')^2}+\frac{D_4'''}{B\,D_4'}-\frac{4\,c^2\,B'}{k\,B\,D_4'}+$$
$$\frac{2\,\Lambda\,D_4\,B'}{B\,D_4'}+\frac{3\,B'\,D_4'}{4\,B^2\,D_4}+\frac{(D_4')^2}{8\,B\,D_4^2}-\frac{B''}{2\,B^2}-\frac{3\,D_4''}{2\,B\,D_4}+\frac{8\,c^2\,D_4''}{k\,(D_4')^2}-\frac{4\,\Lambda\,D_4\,D_4''}{(D_4')^2}-\frac{3\,(D_4'')^2}{2\,B\,(D_4')^2}=\Lambda$$

$$R_{22}: \quad \frac{3\Lambda}{2} + \frac{e^{-\int_{1}^{r}\frac{1}{2}\left(-\frac{4}{\rho}+\frac{B'}{B}-\frac{4B\left(-2c^2+k\Lambda D_4\right)}{k D_4'}-\frac{D_4'}{D_4}-\frac{2D_4''}{D_4'}\right)d\rho}}{r^2 C_1} - \frac{c^2}{k D_4} + \frac{(B')^2}{4B^3} - \frac{8c^4 B}{k^2 (D_4')^2} + \frac{8c^2 B\Lambda D_4}{k (D_4')^2} - \frac{2B\Lambda^2 D_4^2}{(D_4')^2} + \frac{D_4'''}{2B D_4'} -$$

$$\frac{3c^2 B'}{k B D_4'} + \frac{3\Lambda D_4 B'}{2B D_4'} - \frac{B' D_4'}{4B^2 D_4} - \frac{B''}{4B^2} + \frac{D_4''}{2B D_4} + \frac{6c^2 D_4''}{k (D_4')^2} - \frac{3\Lambda D_4 D_4''}{(D_4')^2} + \frac{B' D_4''}{4B^2 D_4'} - \frac{(D_4'')^2}{B (D_4')^2} = \Lambda$$

$$R_{44}: \quad \Lambda = \Lambda$$

However, I was unable to solve these for either B or D.

We next examine various subcases of Case b that add assumptions upfront that could change the outcome of the field equations.

b_Iα: $A(t,r)=a[t]^2$

b_II: $A(t,r)=c^2$

bii: $C(t,r)=1$

biii: $B(t,r)=1$ and $C(t,r)=1$ (if necessary)

b_Iα: $B(t,r)=B(r)$, $C(t,r)=C(r)$, $A(t,r)=a(t)^2$

Restarting Case b from scratch using these constraints, the field equations become:

$$R_{00}: \quad -\frac{2c^2 \dot{a}^2}{a^4} + \frac{2c^2 \ddot{a}}{a^3} = \Lambda$$

$$R_{01}: \quad \frac{\dot{a} D_4'}{a D_4} = 0$$

$$R_{11}: \quad -\frac{2C'}{r C B} + \frac{(C')^2}{2C^2 B} + \frac{B'}{r B^2} + \frac{C' B'}{2C B^2} + \frac{B' D_4'}{2B^2 D_4} + \frac{(D_4')^2}{2B D_4^2} - \frac{C''}{C B} - \frac{D_4''}{B D_4} = \Lambda$$

$$R_{22}: \quad \frac{1}{r^2 C} - \frac{1}{r^2 B} - \frac{2C'}{r C B} + \frac{B'}{2r B^2} + \frac{C' B'}{4C B^2} - \frac{D_4'}{r B D_4} - \frac{C' D_4'}{2C B D_4} - \frac{C''}{2C B} = \Lambda$$

$$R_{44}: \quad \frac{c^2 \ddot{a}}{a^3} - \frac{D_4'}{r B D_4} - \frac{C' D_4'}{2C B D_4} + \frac{B' D_4'}{4B^2 D_4} - \frac{D_4''}{2B D_4} = \Lambda$$

We can solve R_{00} for $a(t)$, and from R_{01} we get that $D_4(r)$ is a constant (because in Case Iα it is assumed that $a(t) \neq$ constant):

$a(t) = -c\,k\,\mathrm{Sec}\left[\frac{\sqrt{\Lambda}\,(k\,t+\upsilon)}{\sqrt{2}}\right]$ where k and υ are constants of integration.

$D_4(r) = K$, a constant, positive because $D_4(r)$ represents a metric

$= 1$, where K is rolled into radius p_4.

With these values, R_{00} and R_{01} become identities, and the remaining equations become

$$R_{11}: \quad -\frac{2C'}{r C B} + \frac{(C')^2}{2C^2 B} + \frac{B'}{r B^2} + \frac{C' B'}{2C B^2} - \frac{C''}{C B} = \Lambda$$

$$R_{22}: \quad \frac{1}{r^2 C} - \frac{1}{r^2 B} - \frac{2 C'}{r C B} + \frac{B'}{2 r B^2} + \frac{C' B'}{4 C B^2} - \frac{C''}{2 C B} = \Lambda$$

$$R_{44}: \quad \frac{3 \Lambda}{4} - \frac{1}{4} \Lambda \operatorname{Cos}\left[\sqrt{2}\ \sqrt{\Lambda}\ (k t + \upsilon)\right] = \Lambda$$

R_{44} can be rewritten as

$$R_{44}: \quad \operatorname{Cos}\left[\sqrt{2}\ \sqrt{\Lambda}\ (k t + \upsilon)\right] = -1$$

$$\Rightarrow \quad \sqrt{2 \Lambda}\,(kt+\upsilon) = -\pi \quad \text{for all } t$$

$$\Rightarrow \quad k=0 \text{ and } \upsilon = -\frac{\pi}{\sqrt{2 \Lambda}}$$

$$\Rightarrow \quad A(t,r) = a(t)^2 = 0, \text{ an unallowable value.}$$

This case has no possible solution.

b_II: $B(t,r)=B(r)$, $C(t,r)=C(r)$, $A(t,r)=c^2$

We try Case b with the additional assumption that $A(t,r)=c^2$. Starting fresh, the field equations are:

$$R_{00}: \quad \frac{2 \ddot{a}}{a} = \Lambda$$

$$R_{01}: \quad \frac{\dot{a} D_4'}{a D_4} = 0$$

$$R_{11}: \quad -\frac{2 C'}{r C B} + \frac{(C')^2}{2 C^2 B} + \frac{B'}{r B^2} + \frac{C' B'}{2 C B^2} + \frac{B' D_4'}{2 B^2 D_4} + \frac{(D_4')^2}{2 B D_4^2} - \frac{C''}{C B} - \frac{D_4''}{B D_4} = \Lambda$$

$$R_{22}: \quad \frac{1}{r^2 C} - \frac{1}{r^2 B} - \frac{2 C'}{r C B} + \frac{B'}{2 r B^2} + \frac{C' B'}{4 C B^2} - \frac{D_4'}{r B D_4} - \frac{C' D_4'}{2 C B D_4} - \frac{C''}{2 C B} = \Lambda$$

$$R_{44}: \quad \frac{\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} - \frac{D_4'}{r B D_4} - \frac{C' D_4'}{2 C B D_4} + \frac{B' D_4'}{4 B^2 D_4} - \frac{D_4''}{2 B D_4} = \Lambda$$

Except for R_{44}, these equations are identical to Case 5D_1A. Thus, from solving R_{00} and R_{01}, respectively, we get

$$a(t) = e^{\frac{t \sqrt{\Lambda}}{\sqrt{2}}} \alpha + e^{-\frac{t \sqrt{\Lambda}}{\sqrt{2}}} \beta \quad \text{where } \alpha \text{ and } \beta \text{ are constants of integration.}$$

$$D_4(r) = 1.$$

Plugging back in, R_{00} and R_{01} are satisfied and can be ignored:

$$R_{11}: \quad -\frac{2 C'}{r C B} + \frac{(C')^2}{2 C^2 B} + \frac{B'}{r B^2} + \frac{C' B'}{2 C B^2} - \frac{C''}{C B} = \Lambda$$

$$R_{22}: \quad \frac{1}{r^2 C} - \frac{1}{r^2 B} - \frac{2 C'}{r C B} + \frac{B'}{2 r B^2} + \frac{C' B'}{4 C B^2} - \frac{C''}{2 C B} = \Lambda$$

$$R_{44}: \quad \frac{e^{2\sqrt{2}\,t\sqrt{\Lambda}}\,\alpha^2\,\Lambda}{\left(e^{\sqrt{2}\,t\sqrt{\Lambda}}\,\alpha+\beta\right)^2}+\frac{\beta^2\,\Lambda}{\left(e^{\sqrt{2}\,t\sqrt{\Lambda}}\,\alpha+\beta\right)^2} = \Lambda$$

Assuming $\Lambda \neq 0$, R_{44} can be simplified by factoring out Λ from both sides:

$$R_{44}: \quad \frac{e^{2\sqrt{2}\,t\sqrt{\Lambda}}\,\alpha^2+\beta^2}{\left(e^{\sqrt{2}\,t\sqrt{\Lambda}}\,\alpha+\beta\right)^2} = 1$$

In order for this equation to be true, we must have

$$\alpha = 0 \text{ or } \beta = 0$$

$$\Rightarrow \quad a(t) = e^{\pm\frac{t\sqrt{\Lambda}}{\sqrt{2}}}\,\alpha.$$

R_{44} has now become an identity.

We can algebraically solve $R_{11}-2R_{22}$ for B:

$$R_{11}-2R_{22}: \quad -\frac{2}{r^2\,C}+\frac{2}{r^2\,B}+\frac{2\,C'}{r\,C\,B}+\frac{(C')^2}{2\,C^2\,B} = -\Lambda$$

$$B(r) = -\frac{(2\,C[r]+r\,C'[r])^2}{2\,C[r]\,\left(-2+r^2\,\Lambda\,C[r]\right)}$$

We see that the remaining equations are both solved:

R_{11}: $\Lambda = \Lambda$

R_{22}: $\Lambda = \Lambda$

We have found an exact solution to the Λ field equations:

$a(t) = e^{\pm\frac{t\sqrt{\Lambda}}{\sqrt{2}}}\,\alpha$ where α is a free parameter,

$A(t,r) = c^2$,

$$B(t,r) = -\frac{(2\,C[r]+r\,C'[r])^2}{2\,C[r]\,\left(-2+r^2\,\Lambda\,C[r]\right)},$$

$\mathbb{C}(t,r) = C[r]$, a free parameter,

$\mathbb{D}_4(t,r) = \mathbb{D}_5(t,r) = a(t)^2$,

and line element

$$ds^2 = c^2\,dt^2 + \frac{(2\,C[r]+r\,C'[r])^2}{2\,C[r]\,\left(-2+r^2\,\Lambda\,C[r]\right)}\,dr^2 - r^2\,C[r]\,d\theta^2 - r^2\,\mathrm{Sin}[\theta]^2\,C[r]\,d\phi^2$$
$$- e^{\sqrt{2}\,t\sqrt{\Lambda}}\,\alpha^2\,p_4^2\,d\psi_4^2$$
$$- e^{\sqrt{2}\,t\sqrt{\Lambda}}\,\alpha^2\,p_5^2\,d\psi_5^2$$

bii: B(t,r)=B(r), C(t,r)=1

We next try a subcase, Case bii, of Case b. Starting fresh, the field equations become

$$R_{00}: \quad \frac{2 c^2 \ddot{a}}{a A} - \frac{c^2 \dot{a} \dot{A}}{a A^2} - \frac{A'}{A r B} + \frac{(A')^2}{4 A^2 B} + \frac{A' B'}{4 A B^2} - \frac{A' D_4'}{2 A B D_4} - \frac{A''}{2 A B} = \Lambda$$

$$R_{01}: \quad -\frac{\dot{a} A'}{a A} + \frac{\dot{a} D_4'}{a D_4} = 0$$

$$R_{11}: \quad \frac{(A')^2}{4 A^2 B} + \frac{B'}{r B^2} + \frac{A' B'}{4 A B^2} + \frac{B' D_4'}{2 B^2 D_4} + \frac{(D_4')^2}{2 B D_4^2} - \frac{A''}{2 A B} - \frac{D_4''}{B D_4} = \Lambda$$

$$R_{22}: \quad \frac{1}{r^2} - \frac{1}{r^2 B} - \frac{A'}{2 A r B} + \frac{B'}{2 r B^2} - \frac{D_4'}{r B D_4} = \Lambda$$

$$R_{44}: \quad \frac{c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} - \frac{c^2 \dot{a} \dot{A}}{2 a A^2} - \frac{D_4'}{r B D_4} - \frac{A' D_4'}{4 A B D_4} + \frac{B' D_4'}{4 B^2 D_4} - \frac{D_4''}{2 B D_4} = \Lambda$$

Like Case b, we can simplify R_{01} by factoring out $\frac{\dot{a}}{a}$, and then solving for A:

$$R_{01}: \quad \frac{A'}{A} - \frac{D_4'}{D_4} = 0$$

$A(t,r) = \mathcal{A}[t]\, D_4[r]$ where $\mathcal{A}(t)$ is a new unknown.

The equations now become

$$R_{00}: \quad \frac{2 c^2 \ddot{a}}{a \mathcal{A} D_4} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{a \mathcal{A}^2 D_4} - \frac{D_4'}{r B D_4} + \frac{B' D_4'}{4 B^2 D_4} - \frac{(D_4')^2}{4 B D_4^2} - \frac{D_4''}{2 B D_4} = \Lambda$$

$$R_{11}: \quad \frac{B'}{r B^2} + \frac{3 B' D_4'}{4 B^2 D_4} + \frac{3 (D_4')^2}{4 B D_4^2} - \frac{3 D_4''}{2 B D_4} = \Lambda$$

$$R_{22}: \quad \frac{1}{r^2} - \frac{1}{r^2 B} + \frac{B'}{2 r B^2} - \frac{3 D_4'}{2 r B D_4} = \Lambda$$

$$R_{44}: \quad \frac{c^2 \dot{a}^2}{a^2 \mathcal{A} D_4} + \frac{c^2 \ddot{a}}{a \mathcal{A} D_4} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2 D_4} - \frac{D_4'}{r B D_4} + \frac{B' D_4'}{4 B^2 D_4} - \frac{(D_4')^2}{4 B D_4^2} - \frac{D_4''}{2 B D_4} = \Lambda$$

From R_{00}-R_{44} we can solve for $\mathcal{A}(t)$:

$$R_{00}-R_{44}: \quad -\frac{c^2 \dot{a}^2}{a^2 \mathcal{A} D_4} + \frac{c^2 \ddot{a}}{a \mathcal{A} D_4} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2 D_4} = 0$$

$$\Rightarrow \quad R_{00}-R_{44}: \quad -2 c^2 \mathcal{A} \dot{a}^2 + 2 a c^2 \mathcal{A} \ddot{a} - a c^2 \dot{a} \dot{\mathcal{A}} = 0$$

$$\mathcal{A}(t) = \frac{k \dot{a}[t]^2}{a[t]^2}$$ where k is a constant of integration.

With this formula, R_{00} and R_{44} are dependent equations, so we can choose to designate R_{44} as redundant.

The remaining equations become:

$$R_{00}: \quad \frac{2\,c^2}{k\,D_4} - \frac{D_4'}{r\,B\,D_4} + \frac{B'\,D_4'}{4\,B^2\,D_4} - \frac{(D_4')^2}{4\,B\,D_4^2} - \frac{D_4''}{2\,B\,D_4} = \Lambda$$

$$R_{11}: \quad \frac{B'}{r\,B^2} + \frac{3\,B'\,D_4'}{4\,B^2\,D_4} + \frac{3\,(D_4')^2}{4\,B\,D_4^2} - \frac{3\,D_4''}{2\,B\,D_4} = \Lambda$$

$$R_{22}: \quad \frac{1}{r^2} - \frac{1}{r^2\,B} + \frac{B'}{2\,r\,B^2} - \frac{3\,D_4'}{2\,r\,B\,D_4} = \Lambda$$

We can solve R_{22} for B:

$$B(r) = \frac{r^2\,D_4^3}{\alpha - \int_1^r \rho\,D_4^2\left(-2\,D_4 + 2\,\Lambda\,\rho^2\,D_4\right)d\rho}$$ where α is a constant of integration.

With R_{22} being an identity, the equations now become

$$R_{00}: \quad \frac{2\,c^2}{k\,D_4} - \frac{\alpha\,D_4'}{2\,r^3\,D_4^4} - \frac{D_4'}{2\,r\,D_4} + \frac{r\,\Lambda\,D_4'}{2\,D_4} + \frac{\alpha\,(D_4')^2}{2\,r^2\,D_4^5} - \frac{\alpha\,D_4''}{2\,r^2\,D_4^4} + \frac{D_4'\int_1^r 2\,\rho\left(-1+\Lambda\,\rho^2\right)D_4^3\,d\rho}{2\,r^3\,D_4^4} - \frac{(D_4')^2\int_1^r 2\,\rho\left(-1+\Lambda\,\rho^2\right)D_4^3\,d\rho}{2\,r^2\,D_4^5} + \frac{D_4''\int_1^r 2\,\rho\left(-1+\Lambda\,\rho^2\right)D_4^3\,d\rho}{2\,r^2\,D_4^4} = \Lambda$$

$$R_{11}: \quad -\frac{2}{r^2} + 2\,\Lambda + \frac{2\,\alpha}{r^4\,D_4^3} + \frac{9\,\alpha\,D_4'}{2\,r^3\,D_4^4} - \frac{3\,D_4'}{2\,r\,D_4} + \frac{3\,r\,\Lambda\,D_4'}{2\,D_4} + \frac{3\,\alpha\,(D_4')^2}{r^2\,D_4^5} - \frac{3\,\alpha\,D_4''}{2\,r^2\,D_4^4} - \frac{2\int_1^r 2\,\rho\left(-1+\Lambda\,\rho^2\right)D_4^3\,d\rho}{r^4\,D_4^3} - \frac{9\,D_4'\int_1^r 2\,\rho\left(-1+\Lambda\,\rho^2\right)D_4^3\,d\rho}{2\,r^3\,D_4^4} - \frac{3\,(D_4')^2\int_1^r 2\,\rho\left(-1+\Lambda\,\rho^2\right)D_4^3\,d\rho}{r^2\,D_4^5} + \frac{3\,D_4''\int_1^r 2\,\rho\left(-1+\Lambda\,\rho^2\right)D_4^3\,d\rho}{2\,r^2\,D_4^4} = \Lambda$$

There is no algebraic combination of the equations that eliminate more than a single integral. However, if we simplify by setting α=0, we can solve R_{00} for the D_4, the remaining unknown. But, the expression is a fraction that includes an integral in an exponent in the numerator and a double integral in the denominator. Checking whether or not the remaining equation, R_{11}, is an identity when using this expression is too computationally intensive for my home computer.

This case, too, is too difficult for me to solve, so we proceed to **Case biii:** $B(r) = 1$, $C(r) = 1$.

biii: B(t,r)=1, C(t,r)=1

Starting fresh again, we have 5 equations in 2 unknowns:

$$R_{00}: \quad \frac{2\,c^2\,\ddot{a}}{a\,A} - \frac{c^2\,\dot{a}\,\dot{A}}{a\,A^2} - \frac{A'}{A\,r} + \frac{(A')^2}{4\,A^2} - \frac{A'\,D_4'}{2\,A\,D_4} - \frac{A''}{2\,A} = \Lambda$$

$$R_{01}: \quad -\frac{\dot{a}\,A'}{a\,A} + \frac{\dot{a}\,D_4'}{a\,D_4} = 0$$

$$R_{11}: \quad \frac{(A')^2}{4\,A^2} + \frac{(D_4')^2}{2\,D_4^2} - \frac{A''}{2\,A} - \frac{D_4''}{D_4} = \Lambda$$

$$R_{22}: \quad -\frac{A'}{2\,A\,r} - \frac{\mathcal{D}_4'}{r\,\mathcal{D}_4} = \Lambda$$

$$R_{44}: \quad \frac{c^2\,\dot{a}^2}{a^2\,A} + \frac{c^2\,\ddot{a}}{a\,A} - \frac{c^2\,\dot{a}\,\dot{A}}{2\,a\,A^2} - \frac{\mathcal{D}_4'}{r\,\mathcal{D}_4} - \frac{A'\,\mathcal{D}_4'}{4\,A\,\mathcal{D}_4} - \frac{\mathcal{D}_4''}{2\,\mathcal{D}_4} = \Lambda$$

Again, like Case b, we can simplify R_{01} by factoring out $\frac{\dot{a}}{a}$, and then solving for A:

$$R_{01}: \quad \frac{A'}{A} - \frac{\mathcal{D}_4'}{\mathcal{D}_4} = 0$$

$A(t,r) = \mathcal{A}[t]\,\mathcal{D}_4[r]$ where $\mathcal{A}(t)$ is a new unknown.

The situation reduces to 4 equations, still in 2 unknowns:

$$R_{00}: \quad \frac{2\,c^2\,\ddot{a}}{a\,\mathcal{A}\,\mathcal{D}_4} - \frac{c^2\,\dot{a}\,\dot{\mathcal{A}}}{a\,\mathcal{A}^2\,\mathcal{D}_4} - \frac{\mathcal{D}_4'}{r\,\mathcal{D}_4} - \frac{(\mathcal{D}_4')^2}{4\,\mathcal{D}_4^2} - \frac{\mathcal{D}_4''}{2\,\mathcal{D}_4} = \Lambda$$

$$R_{11}: \quad \frac{3\,(\mathcal{D}_4')^2}{4\,\mathcal{D}_4^2} - \frac{3\,\mathcal{D}_4''}{2\,\mathcal{D}_4} = \Lambda$$

$$R_{22}: \quad -\frac{3\,\mathcal{D}_4'}{2\,r\,\mathcal{D}_4} = \Lambda$$

$$R_{44}: \quad \frac{c^2\,\dot{a}^2}{a^2\,\mathcal{A}\,\mathcal{D}_4} + \frac{c^2\,\ddot{a}}{a\,\mathcal{A}\,\mathcal{D}_4} - \frac{c^2\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2\,\mathcal{D}_4} - \frac{\mathcal{D}_4'}{r\,\mathcal{D}_4} - \frac{(\mathcal{D}_4')^2}{4\,\mathcal{D}_4^2} - \frac{\mathcal{D}_4''}{2\,\mathcal{D}_4} = \Lambda$$

We can solve R_{22} for $\mathcal{D}_4$ and R_{00}–R_{44} for $\mathcal{A}(t)$:

$$R_{00}-R_{44}: \quad -\frac{c^2\,\dot{a}^2}{a^2\,\mathcal{A}\,\mathcal{D}_4} + \frac{c^2\,\ddot{a}}{a\,\mathcal{A}\,\mathcal{D}_4} - \frac{c^2\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2\,\mathcal{D}_4} = 0$$

$\mathcal{D}_4(r) = e^{-\frac{r^2\,\Lambda}{3}}\,\alpha$ where α is a constant of integration.

$\mathcal{A}(t) = \frac{k\,\dot{a}[t]^2}{a[t]^2}$ where k is a constant of integration.

R_{22} is eliminated, and the remaining equations become:

$$R_{00}: \quad \frac{2\,c^2\,e^{\frac{r^2\,\Lambda}{3}}}{k\,\alpha} + \Lambda - \frac{r^2\,\Lambda^2}{3} = \Lambda$$

$$R_{11}: \quad \Lambda - \frac{r^2\,\Lambda^2}{3} = \Lambda$$

$$R_{44}: \quad \frac{2\,c^2\,e^{\frac{r^2\,\Lambda}{3}}}{k\,\alpha} + \Lambda - \frac{r^2\,\Lambda^2}{3} = \Lambda$$

From R_{11} we get that

$$\Lambda = 0$$

$$\Rightarrow \quad R_{00}: \ \frac{2\,c^2}{k\,\alpha} = 0$$

$\Rightarrow$ c = 0, which is not true.

We conclude that this case has no possible solution.

c: B(t,r)=B(r), $\mathbb{C}$(t,r)=a(t)^2

The field equations become

$$R_{00}: \ \frac{4\,c^2\,\ddot{a}}{a\,A} - \frac{2\,c^2\,\dot{a}\,\dot{A}}{a\,A^2} - \frac{A'}{A\,r\,B} + \frac{(A')^2}{4\,A^2\,B} + \frac{A'\,B'}{4\,A\,B^2} - \frac{A'\,\mathcal{D}_4'}{2\,A\,B\,\mathcal{D}_4} - \frac{A''}{2\,A\,B} = \Lambda$$

$$R_{01}: \ \frac{2\,\dot{a}}{a\,r} - \frac{2\,\dot{a}\,A'}{a\,A} + \frac{\dot{a}\,\mathcal{D}_4'}{a\,\mathcal{D}_4} = 0$$

$$R_{11}: \ \frac{(A')^2}{4\,A^2\,B} + \frac{B'}{r\,B^2} + \frac{A'\,B'}{4\,A\,B^2} + \frac{B'\,\mathcal{D}_4'}{2\,B^2\,\mathcal{D}_4} + \frac{(\mathcal{D}_4')^2}{2\,B\,\mathcal{D}_4^2} - \frac{A''}{2\,A\,B} - \frac{\mathcal{D}_4''}{B\,\mathcal{D}_4} = \Lambda$$

$$R_{22}: \ \frac{1}{a^2\,r^2} - \frac{1}{r^2\,B} + \frac{3\,c^2\,\dot{a}^2}{a^2\,A} + \frac{c^2\,\ddot{a}}{a\,A} - \frac{c^2\,\dot{a}\,\dot{A}}{2\,a\,A^2} - \frac{A'}{2\,A\,r\,B} + \frac{B'}{2\,r\,B^2} - \frac{\mathcal{D}_4'}{r\,B\,\mathcal{D}_4} = \Lambda$$

$$R_{44}: \ \frac{3\,c^2\,\dot{a}^2}{a^2\,A} + \frac{c^2\,\ddot{a}}{a\,A} - \frac{c^2\,\dot{a}\,\dot{A}}{2\,a\,A^2} - \frac{\mathcal{D}_4'}{r\,B\,\mathcal{D}_4} - \frac{A'\,\mathcal{D}_4'}{4\,A\,B\,\mathcal{D}_4} + \frac{B'\,\mathcal{D}_4'}{4\,B^2\,\mathcal{D}_4} - \frac{\mathcal{D}_4''}{2\,B\,\mathcal{D}_4} = \Lambda$$

We can simplify R_{01} by factoring out $\frac{\dot{a}}{a}$, and then solving for A:

$$R_{01}: \ -\frac{2}{r} + \frac{2\,A'}{A} - \frac{\mathcal{D}_4'}{\mathcal{D}_4} = 0$$

$A(t,r) = r\,\mathcal{A}[t]\,\sqrt{\mathcal{D}_4[r]}$ where $\mathcal{A}(t)$ is a new unknown.

This eliminates R_{01}, and the remaining 4 equations become

$$R_{00}: \ -\frac{3}{4\,r^2\,B} + \frac{4\,c^2\,\ddot{a}}{a\,r\,\mathcal{A}\,\sqrt{\mathcal{D}_4}} - \frac{2\,c^2\,\dot{a}\,\dot{\mathcal{A}}}{a\,r\,\mathcal{A}^2\,\sqrt{\mathcal{D}_4}} + \frac{B'}{4\,r\,B^2} - \frac{5\,\mathcal{D}_4'}{4\,r\,B\,\mathcal{D}_4} + \frac{B'\,\mathcal{D}_4'}{8\,B^2\,\mathcal{D}_4} - \frac{(\mathcal{D}_4')^2}{16\,B\,\mathcal{D}_4^2} - \frac{\mathcal{D}_4''}{4\,B\,\mathcal{D}_4} = \Lambda$$

$$R_{11}: \ \frac{1}{4\,r^2\,B} + \frac{5\,B'}{4\,r\,B^2} - \frac{\mathcal{D}_4'}{4\,r\,B\,\mathcal{D}_4} + \frac{5\,B'\,\mathcal{D}_4'}{8\,B^2\,\mathcal{D}_4} + \frac{11\,(\mathcal{D}_4')^2}{16\,B\,\mathcal{D}_4^2} - \frac{5\,\mathcal{D}_4''}{4\,B\,\mathcal{D}_4} = \Lambda$$

$$R_{22}: \ \frac{1}{a^2\,r^2} - \frac{3}{2\,r^2\,B} + \frac{3\,c^2\,\dot{a}^2}{a^2\,r\,\mathcal{A}\,\sqrt{\mathcal{D}_4}} + \frac{c^2\,\ddot{a}}{a\,r\,\mathcal{A}\,\sqrt{\mathcal{D}_4}} - \frac{c^2\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,r\,\mathcal{A}^2\,\sqrt{\mathcal{D}_4}} + \frac{B'}{2\,r\,B^2} - \frac{5\,\mathcal{D}_4'}{4\,r\,B\,\mathcal{D}_4} = \Lambda$$

$$R_{44}: \ \frac{3\,c^2\,\dot{a}^2}{a^2\,r\,\mathcal{A}\,\sqrt{\mathcal{D}_4}} + \frac{c^2\,\ddot{a}}{a\,r\,\mathcal{A}\,\sqrt{\mathcal{D}_4}} - \frac{c^2\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,r\,\mathcal{A}^2\,\sqrt{\mathcal{D}_4}} - \frac{5\,\mathcal{D}_4'}{4\,r\,B\,\mathcal{D}_4} + \frac{B'\,\mathcal{D}_4'}{4\,B^2\,\mathcal{D}_4} - \frac{(\mathcal{D}_4')^2}{8\,B\,\mathcal{D}_4^2} - \frac{\mathcal{D}_4''}{2\,B\,\mathcal{D}_4} = \Lambda$$

Now we can solve $2R_{00}-R_{22}-R_{44}$ for $\mathcal{A}$:

$$2R_{00}-R_{22}-R_{44}: \ -\frac{1}{a^2\,r^2} - \frac{6\,c^2\,\dot{a}^2}{a^2\,r\,\mathcal{A}\,\sqrt{\mathcal{D}_4}} + \frac{6\,c^2\,\ddot{a}}{a\,r\,\mathcal{A}\,\sqrt{\mathcal{D}_4}} - \frac{3\,c^2\,\dot{a}\,\dot{\mathcal{A}}}{a\,r\,\mathcal{A}^2\,\sqrt{\mathcal{D}_4}} = 0$$

$$\mathcal{A}(t) = \frac{6\,c^2\,r\,\dot{a}[t]^2}{6\,c^2\,r\,\alpha\,a[t]^2 - \sqrt{\mathcal{D}_4[r]}}$$ where α is a constant of integration;

This formula for $\mathcal{A}(t)$ makes R_{00}, R_{22},
and R_{44} dependent, so we are free to declare R_{44} redundant.

Also, in order for $\mathcal{A}(t)$ to be strictly a function of t, $\sqrt{\mathcal{D}_4}$ must be proportional to r:

$\mathcal{D}_4 = K r^2$ where K is the proportionality constant.

$$\Rightarrow \quad \mathcal{A}(t) = -\frac{6\,c^2\,\dot{a}[t]^2}{\sqrt{K} - 6\,c^2\,\alpha\,a[t]^2}$$

Plugging $\mathcal{A}(t)$ and $\mathcal{D}_4(r)$ back in, the equations become:

R_{00}: $$-\frac{4}{r^2\,B} + \frac{4\,c^2\,\alpha}{r^2\,\sqrt{K}} + \frac{B'}{2\,r\,B^2} = \Lambda$$

R_{11}: $$\frac{5\,B'}{2\,r\,B^2} = \Lambda$$

R_{22}: $$\frac{1}{2\,a^2\,r^2} - \frac{4}{r^2\,B} + \frac{4\,c^2\,\alpha}{r^2\,\sqrt{K}} + \frac{B'}{2\,r\,B^2} = \Lambda$$

Solving R_{11} for B yields

$$B(r) = -\frac{5}{\beta + r^2\,\Lambda}$$ where β is a constant of integration.

Plugging this into the remaining 2 equations yields:

R_{00}: $$\frac{4\,\beta}{5\,r^2} + \frac{4\,c^2\,\alpha}{r^2\,\sqrt{K}} + \Lambda = \Lambda$$

R_{22}: $$\frac{4\,\beta}{5} + \frac{4\,c^2\,\alpha}{\sqrt{K}} + \frac{1}{2\,a[t]^2} = 0$$

We can algebraically solve R_{22} for $a(t)$:

$$a(t) = \frac{\sqrt{\frac{5}{2}}\,K^{1/4}}{2\,\sqrt{-5\,c^2\,\alpha - \beta\,\sqrt{K}}}$$, a constant.

However, this makes

$\mathcal{A}(t) = 0$,

an unallowable value.

This case has no possible solution.

We next re-do this case, but with various additional assumptions. By making the assumptions upfront, before the ODEs have been solved, it may be possible to obtain solutions.

Iα: A(t,r) = $\mathcal{A}$(t)

Iβ: A(t,r) = A(r)

II: A(t,r) = c^2

c_Iα: B(t,r)=B(r), $\mathbb{C}$(t,r)=a(t)^2, A(t,r)=$\mathcal{A}$(t)

Starting fresh, the field equations are:

$$R_{00}: \quad \frac{4 c^2 \ddot{a}}{a \mathcal{A}} - \frac{2 c^2 \dot{a} \dot{\mathcal{A}}}{a \mathcal{A}^2} = \Lambda$$

$$R_{01}: \quad \frac{2 \dot{a}}{a r} + \frac{\dot{a} \mathbb{D}_4'}{a \mathbb{D}_4} = 0$$

$$R_{11}: \quad \frac{B'}{r B^2} + \frac{B' \mathbb{D}_4'}{2 B^2 \mathbb{D}_4} + \frac{(\mathbb{D}_4')^2}{2 B \mathbb{D}_4^2} - \frac{\mathbb{D}_4''}{B \mathbb{D}_4} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2 r^2} - \frac{1}{r^2 B} + \frac{3 c^2 \dot{a}^2}{a^2 \mathcal{A}} + \frac{c^2 \ddot{a}}{a \mathcal{A}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2} + \frac{B'}{2 r B^2} - \frac{\mathbb{D}_4'}{r B \mathbb{D}_4} = \Lambda$$

$$R_{44}: \quad \frac{3 c^2 \dot{a}^2}{a^2 \mathcal{A}} + \frac{c^2 \ddot{a}}{a \mathcal{A}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2} - \frac{\mathbb{D}_4'}{r B \mathbb{D}_4} + \frac{B' \mathbb{D}_4'}{4 B^2 \mathbb{D}_4} - \frac{\mathbb{D}_4''}{2 B \mathbb{D}_4} = \Lambda$$

We can simplify R_{01} by factoring out $\frac{\dot{a}}{a}$, and then solving for $\mathbb{D}_4$:

$$R_{01}: \quad -\frac{2}{r} - \frac{\mathbb{D}_4'}{\mathbb{D}_4} = 0$$

$\mathbb{D}_4(r) = \frac{k}{r^2}$ where k is a constant of integration

whose positive part can be rolled into p_4,

$$\mathbb{D}_4(r) = \pm\frac{1}{r^2}.$$

This eliminates R_{01}, and the remaining 4 equations become

$$R_{00}: \quad \frac{4 c^2 \ddot{a}}{a \mathcal{A}} - \frac{2 c^2 \dot{a} \dot{\mathcal{A}}}{a \mathcal{A}^2} = \Lambda$$

$$R_{11}: \quad -\frac{4}{r^2 B} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2 r^2} + \frac{1}{r^2 B} + \frac{3 c^2 \dot{a}^2}{a^2 \mathcal{A}} + \frac{c^2 \ddot{a}}{a \mathcal{A}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2} + \frac{B'}{2 r B^2} = \Lambda$$

$$R_{44}: \quad -\frac{1}{r^2 B} + \frac{3 c^2 \dot{a}^2}{a^2 \mathcal{A}} + \frac{c^2 \ddot{a}}{a \mathcal{A}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2} - \frac{B'}{2 r B^2} = \Lambda$$

We can algebraically solve R_{11} for B:

$$B(r) = -\frac{4}{r^2 \Lambda}.$$

R_{11} is solved, and the remaining equations become:

$$R_{00}: \quad \frac{4 c^2 \ddot{a}}{a \mathcal{A}} - \frac{2 c^2 \dot{a} \dot{\mathcal{A}}}{a \mathcal{A}^2} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2 r^2} + \frac{3 c^2 \dot{a}^2}{a^2 \mathcal{A}} + \frac{c^2 \ddot{a}}{a \mathcal{A}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2} = \Lambda$$

$$R_{44}: \quad \frac{3 c^2 \dot{a}^2}{a^2 \mathcal{A}} + \frac{c^2 \ddot{a}}{a \mathcal{A}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2} = \Lambda$$

$R_{22}-R_{44}$ has no solution.

$$R_{22}-R_{44}: \quad \frac{1}{r^2 a[t]^2} = 0$$

Case c_1α has no possible solution.

c_Iβ: B(t,r)=B(r), C(t,r)=a(t)^2, A(t,r)=A(r)

Starting fresh, the field equations are identical to those in Case c except that all instances of A(t,r) are replaced by A(r):

$$R_{00}: \quad \frac{4 c^2 \ddot{a}}{a A} - \frac{A'}{r A B} + \frac{(A')^2}{4 A^2 B} + \frac{A' B'}{4 A B^2} - \frac{A' D_4'}{2 A B D_4} - \frac{A''}{2 A B} = \Lambda$$

$$R_{01}: \quad \frac{2 \dot{a}}{a r} - \frac{2 \dot{a} A'}{a A} + \frac{\dot{a} D_4'}{a D_4} = 0$$

$$R_{11}: \quad \frac{(A')^2}{4 A^2 B} + \frac{B'}{r B^2} + \frac{A' B'}{4 A B^2} + \frac{B' D_4'}{2 B^2 D_4} + \frac{(D_4')^2}{2 B D_4^2} - \frac{A''}{2 A B} - \frac{D_4''}{B D_4} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2 r^2} - \frac{1}{r^2 B} + \frac{3 c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} - \frac{A'}{2 r A B} + \frac{B'}{2 r B^2} - \frac{D_4'}{r B D_4} = \Lambda$$

$$R_{44}: \quad \frac{3 c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} - \frac{D_4'}{r B D_4} - \frac{A' D_4'}{4 A B D_4} + \frac{B' D_4'}{4 B^2 D_4} - \frac{D_4''}{2 B D_4} = \Lambda$$

We can simplify R_{01} by factoring out $\frac{\dot{a}}{a}$, and then solving for D_4:

$$R_{01}: \quad -\frac{2}{r} + \frac{2 A'}{A} - \frac{D_4'}{D_4} = 0$$

$$\mathcal{D}_4(r) = \frac{k\,A[r]^2}{r^2}$$ where k is a constant of integration

whose positive part can be rolled into p_4,

$$\mathcal{D}_4(r) = \pm\frac{A[r]^2}{r^2}.$$

This eliminates R_{01}, and the remaining 4 equations become

$$R_{00}:\quad \frac{4\,c^2\,\ddot{a}}{a\,A} - \frac{3\,(A')^2}{4\,A^2\,B} + \frac{A'\,B'}{4\,A\,B^2} - \frac{A''}{2\,A\,B} = \Lambda$$

$$R_{11}:\quad -\frac{4}{r^2\,B} + \frac{4\,A'}{r\,A\,B} + \frac{(A')^2}{4\,A^2\,B} + \frac{5\,A'\,B'}{4\,A\,B^2} - \frac{5\,A''}{2\,A\,B} = \Lambda$$

$$R_{22}:\quad \frac{1}{a^2\,r^2} + \frac{1}{r^2\,B} + \frac{3\,c^2\,\dot{a}^2}{a^2\,A} + \frac{c^2\,\ddot{a}}{a\,A} - \frac{5\,A'}{2\,r\,A\,B} + \frac{B'}{2\,r\,B^2} = \Lambda$$

$$R_{44}:\quad -\frac{1}{r^2\,B} + \frac{3\,c^2\,\dot{a}^2}{a^2\,A} + \frac{c^2\,\ddot{a}}{a\,A} + \frac{5\,A'}{2\,r\,A\,B} - \frac{3\,(A')^2}{2\,A^2\,B} - \frac{B'}{2\,r\,B^2} + \frac{A'\,B'}{2\,A\,B^2} - \frac{A''}{A\,B} = \Lambda$$

Now we can algebraically solve for A:

$$2R_{00}-R_{22}-R_{44}:\quad -\frac{1}{a^2\,r^2} - \frac{6\,c^2\,\dot{a}^2}{a^2\,A} + \frac{6\,c^2\,\ddot{a}}{a\,A} = 0$$

$$\Rightarrow\quad A(r) = -6\,c^2\,r^2\left(\dot{a}[t]^2 - a[t]\,\ddot{a}[t]\right)$$

In order for $A(r)$ to be strictly a function of r,
the expression in the parentheses must equal a constant, say K:

$$\Rightarrow\quad A(r) = -6\,c^2\,r^2\,K$$

Solving 'parentheses = K' for $a(t)$ yields

$$a(t) = -\frac{\sqrt{K}\,\mathrm{Sinh}[t\,\alpha+\beta]}{\alpha}$$ where α, β, and K are constants

$$\Rightarrow\quad \mathcal{D}_4(r) = \pm 36\,c^4\,r^2\,K^2$$

Temporarily letting $k = -6c^2K$ simplifies $A(r)$ to

$$\Rightarrow\quad A(r) = k\,r^2$$

Having solved $2R_{00}-R_{22}-R_{44}=0$, the equations R_{00}, R_{22},
and R_{44} are dependent and we are free to designate R_{44} as redundant.

Plugging in $A(r)$ (and, for now, expressing
$a(t)$ with just its symbol), the remaining 3 equations become:

R_{00}: $-\frac{4}{r^2 B} + \frac{4 c^2 \ddot{a}}{a k r^2} + \frac{B'}{2 r B^2} = \Lambda$

R_{11}: $\frac{5 B'}{2 r B^2} = \Lambda$

R_{22}: $\frac{1}{a^2 r^2} - \frac{4}{r^2 B} + \frac{3 c^2 \dot{a}^2}{a^2 k r^2} + \frac{c^2 \ddot{a}}{a k r^2} + \frac{B'}{2 r B^2} = \Lambda$

From R_{11}, we see that $\frac{B'}{2 r B^2} = \frac{\Lambda}{5}$.

Plugging this in, we simplify R_{00} and R_{22}:

R_{00}: $-\frac{4}{r^2 B} + \frac{\Lambda}{5} + \frac{4 c^2 \ddot{a}}{a k r^2} = \Lambda$

R_{22}: $\frac{1}{a^2 r^2} - \frac{4}{r^2 B} + \frac{\Lambda}{5} + \frac{3 c^2 \dot{a}^2}{a^2 k r^2} + \frac{c^2 \ddot{a}}{a k r^2} = \Lambda$

Solving R_{00} gives us

$$B(r) = -\frac{5 k a[t]}{k r^2 \Lambda a[t] - 5 c^2 \ddot{a}[t]}$$

When we plug in the formula for $a(t)$, we see that B indeed is a function solely of r:

$$B(r) = \frac{5 k}{5 c^2 \alpha^2 - k r^2 \Lambda}$$

This satisifies R_{11} as well as R_{00}:

R_{00}: $\Lambda = \Lambda$

R_{11}: $\Lambda = \Lambda$

R_{22}: $\frac{\left(-6 c^2 \alpha^2 - k r^2 \Lambda + k r^2 \Lambda \operatorname{Cosh}[2 (t \alpha + \beta)]\right) \operatorname{Csch}[t \alpha + \beta]^2}{2 k r^2} = \Lambda$

When $\beta=0$, there is no value of α that satisfies R_{22}.

When $\beta\neq0$, the only value of α that satisfies R_{22} is $\alpha=0$:

R_{22}: $-\frac{1}{2} \Lambda \operatorname{Csch}[\beta]^2 + \frac{1}{2} \Lambda \operatorname{Cosh}[2 \beta] \operatorname{Csch}[\beta]^2 = \Lambda$

where $\beta\neq0$ or else $\operatorname{Csch}(\beta)$ is undefined,

R_{22}: $\frac{1}{2} \Lambda (-1 + \operatorname{Cosh}[2 \beta]) \operatorname{Csch}[\beta]^2 = \Lambda$

Since $\operatorname{Cosh}(2\beta) = \operatorname{Cosh}[\beta]^2 + \operatorname{Sinh}[\beta]^2$, $-1 + \operatorname{Cosh}(2\beta) = 2\operatorname{Sinh}^2(\beta)$, and

R_{22}: $\Lambda = \Lambda$ ✓

However, using L'Hospital's Rule

$$a(t) = -\lim_{\alpha\to 0} \frac{\sqrt{K}\,\mathrm{Sinh}[t\,\alpha+\beta]}{\alpha} = \begin{cases} \sqrt{K}\,t & \text{if } \beta = 0 \\ \text{Indeterminate} & \text{otherwise} \end{cases}$$

We have discovered that the only solution
to R_{22} requires $\beta\neq 0$, and that causes $a(t)$ to be undefined.

Case c_Iβ has no possible solution.

c_II: $B(t,r)=B(r)$, $\mathbb{C}(t,r)=a(t)^2$, $A(t,r)=c^2$

Starting fresh, the field equations are:

$$R_{00}:\quad \frac{4\,\ddot{a}}{a} = \Lambda$$

$$R_{01}:\quad \frac{2\,\dot{a}}{a\,r} + \frac{\dot{a}\,\mathbb{D}_4'}{a\,\mathbb{D}_4} = 0$$

$$R_{11}:\quad \frac{B'}{r\,B^2} + \frac{B'\,\mathbb{D}_4'}{2\,B^2\,\mathbb{D}_4} + \frac{(\mathbb{D}_4')^2}{2\,B\,\mathbb{D}_4^2} - \frac{\mathbb{D}_4''}{B\,\mathbb{D}_4} = \Lambda$$

$$R_{22}:\quad \frac{1}{a^2\,r^2} - \frac{1}{r^2\,B} + \frac{3\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} + \frac{B'}{2\,r\,B^2} - \frac{\mathbb{D}_4'}{r\,B\,\mathbb{D}_4} = \Lambda$$

$$R_{44}:\quad \frac{3\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} - \frac{\mathbb{D}_4'}{r\,B\,\mathbb{D}_4} + \frac{B'\,\mathbb{D}_4'}{4\,B^2\,\mathbb{D}_4} - \frac{\mathbb{D}_4''}{2\,B\,\mathbb{D}_4} = \Lambda$$

Solving R_{00} yields

$$a(t) = e^{\frac{t\sqrt{\Lambda}}{2}}\,\alpha + e^{-\frac{t\sqrt{\Lambda}}{2}}\,\beta.$$

We don't actually use this formula. We will
continue to work with generic $a(t)$, noting that R_{00} is an identity.

We can simplify R_{01} by factoring out $\frac{\dot{a}}{a}$, and then solving for $\mathbb{D}_5$:

$$R_{01}:\quad -\frac{2}{r} - \frac{\mathbb{D}_4'}{\mathbb{D}_4} = 0$$

$$\mathbb{D}_4(r) = \frac{K}{r^2} \quad \text{where } K \text{ is a constant of integration}$$

R_{00} and R_{01} are identities, and the remaining 4 equations become

$$R_{11}:\quad -\frac{4}{r^2\,B} = \Lambda$$

$$R_{22}:\quad \frac{1}{a^2\,r^2} + \frac{1}{r^2\,B} + \frac{3\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} + \frac{B'}{2\,r\,B^2} = \Lambda$$

$$R_{44}: \quad -\frac{1}{r^2 B} + \frac{3\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} - \frac{B'}{2 r B^2} = \Lambda$$

We can solve R_{11} for B:

$$B(r) = -\frac{4}{r^2 \Lambda}.$$

The resulting equations are:

$$R_{22}: \quad \frac{1}{a^2 r^2} + \frac{3\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} = \Lambda$$

$$R_{44}: \quad \frac{3\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} = \Lambda$$

We see that $R_{22}-R_{44}$ has no solution:

$$R_{22}-R_{44}: \quad \frac{1}{r^2 a[t]^2} = 0$$

d: $B(t,r)=a(t)^2$, $\mathbb{C}(t,r)=C(r)$

We preface this case with an observation of what is to come. We will generate 3 distinct solutions, Cases d, d_Iα, and d_II, that are very similar. The solutions differ only in $a(t)$ and $A(t,r)$. $B(r)$ and $\mathbb{C}(r)$ are identical and the $D_4(r)$'s are essentially the same. These solutions are particularly difficult to develop.

We will also generate the subcase of Case d having $g_{00}=1$.`

The field equations for Case d become

$$R_{00}: \quad \frac{3 c^2 \ddot{a}}{a A} - \frac{3 c^2 \dot{a} \dot{A}}{2 a A^2} - \frac{A'}{a^2 A r} + \frac{(A')^2}{4 a^2 A^2} - \frac{A' C'}{2 a^2 A C} - \frac{A' D_4'}{2 a^2 A D_4} - \frac{A''}{2 a^2 A} = \Lambda$$

$$R_{01}: \quad -\frac{2\dot{a}}{a r} - \frac{\dot{a} A'}{a A} - \frac{\dot{a} C'}{a C} = 0$$

$$R_{11}: \quad \frac{2 c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} - \frac{c^2 \dot{a} \dot{A}}{2 a A^2} + \frac{(A')^2}{4 a^2 A^2} - \frac{2 C'}{a^2 r C} + \frac{(C')^2}{2 a^2 C^2} + \frac{(D_4')^2}{2 a^2 D_4^2} - \frac{A''}{2 a^2 A} - \frac{C''}{a^2 C} - \frac{D_4''}{a^2 D_4} = \Lambda$$

$$R_{22}: \quad -\frac{1}{a^2 r^2} + \frac{1}{r^2 C} - \frac{A'}{2 a^2 A r} - \frac{2 C'}{a^2 r C} - \frac{A' C'}{4 a^2 A C} - \frac{D_4'}{a^2 r D_4} - \frac{C' D_4'}{2 a^2 C D_4} - \frac{C''}{2 a^2 C} = \Lambda$$

$$R_{44}: \quad \frac{2 c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} - \frac{c^2 \dot{a} \dot{A}}{2 a A^2} - \frac{D_4'}{a^2 r D_4} - \frac{A' D_4'}{4 a^2 A D_4} - \frac{C' D_4'}{2 a^2 C D_4} - \frac{D_4''}{2 a^2 D_4} = \Lambda$$

We can simplify R_{01} by factoring out $\frac{\dot{a}}{a}$, and then solving for A:

$$R_{01}: \quad \frac{2}{r} + \frac{A'}{A} + \frac{C'}{C} = 0$$

$$A(t,r) = \frac{\mathcal{A}[t]}{r^2\, C[r]} \quad \text{where } \mathcal{A}(t) \text{ is a new unknown.}$$

This eliminates R_{01}, and the remaining equations become

$$R_{00}: \quad \frac{3\,c^2\,r^2\,C\,\ddot{a}}{a\,\mathcal{A}} - \frac{3\,c^2\,r^2\,C\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2} + \frac{C'}{a^2\,r\,C} - \frac{(C')^2}{4\,a^2\,C^2} + \frac{D_4'}{a^2\,r\,D_4} + \frac{C'\,D_4'}{2\,a^2\,C\,D_4} + \frac{C''}{2\,a^2\,C} = \Lambda$$

$$R_{11}: \quad -\frac{2}{a^2\,r^2} + \frac{2\,c^2\,r^2\,C\,\dot{a}^2}{a^2\,\mathcal{A}} + \frac{c^2\,r^2\,C\,\ddot{a}}{a\,\mathcal{A}} - \frac{c^2\,r^2\,C\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2} - \frac{3\,C'}{a^2\,r\,C} - \frac{(C')^2}{4\,a^2\,C^2} + \frac{(D_4')^2}{2\,a^2\,D_4^2} - \frac{C''}{2\,a^2\,C} - \frac{D_4''}{a^2\,D_4} = \Lambda$$

$$R_{22}: \quad \frac{1}{r^2\,C} - \frac{C'}{a^2\,r\,C} + \frac{(C')^2}{4\,a^2\,C^2} - \frac{D_4'}{a^2\,r\,D_4} - \frac{C'\,D_4'}{2\,a^2\,C\,D_4} - \frac{C''}{2\,a^2\,C} = \Lambda$$

$$R_{44}: \quad \frac{2\,c^2\,r^2\,C\,\dot{a}^2}{a^2\,\mathcal{A}} + \frac{c^2\,r^2\,C\,\ddot{a}}{a\,\mathcal{A}} - \frac{c^2\,r^2\,C\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2} - \frac{D_4'}{2\,a^2\,r\,D_4} - \frac{C'\,D_4'}{4\,a^2\,C\,D_4} - \frac{D_4''}{2\,a^2\,D_4} = \Lambda$$

We can solve $R_{00}+R_{22}$ for $\mathcal{A}$:

$$R_{00}+R_{22}: \quad \frac{1}{r^2\,C} + \frac{3\,c^2\,r^2\,C\,\ddot{a}}{a\,\mathcal{A}} - \frac{3\,c^2\,r^2\,C\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2} = 2\,\Lambda$$

$$\mathcal{A}(t) = \frac{3\,c^2\,r^4\,C[r]^2\,\dot{a}[t]^2}{-a[t]^2 + 2\,r^2\,\Lambda\,a[t]^2\,C[r] + 3\,c^2\,r^4\,\kappa\,C[r]^2}$$

$$= \frac{3\,c^2\,\dot{a}[t]^2}{3\,c^2\,\kappa + a[t]^2\left(-\frac{1}{r^4\,C[r]^2} + \frac{2\,\Lambda}{r^2\,C[r]}\right)}$$

where κ is a constant of integration.

In order for $\mathcal{A}(t)$ not to be a function of r, the term in parentheses must be a constant.

$$\Rightarrow \quad C(r) = \frac{K}{r^2} \quad \text{for some proportionality constant } K,$$

$$\Rightarrow \quad \mathcal{A}(t) = \frac{3\,c^2\,K^2\,\dot{a}[t]^2}{3\,c^2\,\kappa\,K^2 - a[t]^2 + 2\,K\,\Lambda\,a[t]^2}.$$

Moreover, this makes $A(t,r)$ strictly a function of t:

$$A(t,r) = \frac{\mathcal{A}[t]}{r^2\,C[r]} = \frac{\mathcal{A}(t)}{K}.$$

The equations now become

$$R_{00}: \quad -\frac{1}{K} + 2\,\Lambda = \Lambda$$

$$R_{11}: \quad -\frac{1}{K} + \frac{2\,c^2\,\kappa\,K}{a^2} + 2\,\Lambda + \frac{(D_4')^2}{2\,a^2\,D_4^2} - \frac{D_4''}{a^2\,D_4} = \Lambda$$

R_{22}: $\frac{1}{K} = \Lambda$

R_{44}: $-\frac{1}{K} + \frac{2 c^2 \kappa K}{a^2} + 2 \Lambda - \frac{\mathcal{D}_4''}{2 a^2 \mathcal{D}_4} = \Lambda$

From R_{22} we get

$\Lambda \neq 0$ and $K = \frac{1}{\Lambda}$.

This modifies A(t,r) again:

$$A(t,r) = \Lambda\, \mathcal{A}(t) = (\Lambda) \left(\frac{3 c^2 \dot{a}[t]^2}{3 c^2 \kappa + \Lambda^2 a[t]^2}\right) =$$

R_{00} and R_{22} have become identities, and the other equations become

R_{11}: $\frac{2 c^2 \kappa}{a^2 \Lambda} + \frac{(\mathcal{D}_4')^2}{2 a^2 \mathcal{D}_4^2} - \frac{\mathcal{D}_4''}{a^2 \mathcal{D}_4} = 0$

R_{44}: $\frac{2 c^2 \kappa}{a^2 \Lambda} - \frac{\mathcal{D}_4''}{2 a^2 \mathcal{D}_4} = 0$

We can solve $R_{11}-2R_{44}$ for $\mathcal{D}_4$:

$R_{11}-2R_{44}$: $-\frac{2 c^2 \kappa}{a^2 \Lambda} - \Lambda + \frac{(\mathcal{D}_4')^2}{2 a^2 \mathcal{D}_4^2} = -\Lambda$

$\mathcal{D}_4(r) = e^{\pm \frac{2 c r \sqrt{\kappa}}{\sqrt{\Lambda}}} \gamma$ where γ is a constant of integration.

Since $g_{44} = -a(t)^2 \mathcal{D}_4(r) p_4{}^2$, $\gamma p_4{}^2$ equals $\hat{p}_4{}^2$ for
a different radius, $\hat{p}_4$. So, we can roll γ into the radius p_4:

$\mathcal{D}_4(r) = e^{\pm \frac{2 c r \sqrt{\kappa}}{\sqrt{\Lambda}}}$

Plugging this in, the remaining equations are satisfied:

R_{11}: $0 = 0$

R_{44}: $0 = 0$

All of the equations are satisfied,
so we have an exact solution to the $\Lambda \neq 0$ field equations:

a(t) is a free parameter,

$A(t,r) = \frac{3 c^2 \Lambda \dot{a}[t]^2}{3 c^2 \kappa + \Lambda^2 a[t]^2}$ where κ is a free constant,

$B(t,r) = a[t]^2$,

$$\mathbb{C}(t,r) = \frac{1}{r^2\,\Lambda} \quad \text{where } \Lambda\neq 0,$$

$$\mathbb{D}_4(t,r) = \mathbb{D}_5(t,r) = a(t)^2\mathcal{D}_4(r) = e^{\pm\frac{2\,c\,r\,\sqrt{\kappa}}{\sqrt{\Lambda}}}\,a[t]^2,$$

and line element

$$ds^2 = \frac{3\,c^2\,\Lambda\,\dot{a}[t]^2}{3\,c^2\,\kappa+\Lambda^2\,a[t]^2}\,dt^2 - a[t]^2\,dr^2 - \frac{1}{\Lambda}\,d\theta^2 - \frac{\mathrm{Sin}[\theta]^2}{\Lambda}\,d\phi^2 - e^{\frac{2\,c\,r\,\sqrt{\kappa}}{\sqrt{\Lambda}}}\,a[t]^2\,p_4^2\,d\psi_4^2 - e^{\frac{2\,c\,r\,\sqrt{\kappa}}{\sqrt{\Lambda}}}\,a[t]^2\,p_5^2\,d\psi_5^2$$

d-II: Solve $A(t,r)=c^2$ for $a(t)$

Solving $\dfrac{3\,c^2\,\Lambda\,\dot{a}[t]^2}{3\,c^2\,\kappa+\Lambda^2\,a[t]^2} = c^2$ yields

$$a(t) = -\frac{\sqrt{3}\,c\,\sqrt{\kappa}\,\mathrm{Sinh}\left[\frac{\sqrt{\Lambda}\,(\upsilon\pm t)}{\sqrt{3}}\right]}{\Lambda} \quad \text{where } \upsilon \text{ is a free constant.}$$

d_Iα: $B(t,r)=a(t)^2$, $\mathbb{C}(t,r)=\mathbb{C}(r)$, $A(t,r)=A(r)$

As usual, when we re-do a case with an additional assumption, we could obtain another solution. In this case, we do obtain a new solution. We are re-doing Case d with the assumption that $\mathcal{A}(t) = 1$; i.e., $A(t,r) = \mathcal{A}(t)\,A(r) = A(r)$.

Starting fresh, the field equations become

$$R_{00}:\quad \frac{3\,c^2\,\ddot{a}}{a\,A} - \frac{A'}{a^2\,r\,A} - \frac{\mathbb{C}'\,A'}{2\,a^2\,\mathbb{C}\,A} + \frac{(A')^2}{4\,a^2\,A^2} - \frac{A'\,\mathcal{D}_4'}{2\,a^2\,A\,\mathcal{D}_4} - \frac{A''}{2\,a^2\,A} = \Lambda$$

$$R_{01}:\quad -\frac{2\,\dot{a}}{a\,r} - \frac{\dot{a}\,\mathbb{C}'}{a\,\mathbb{C}} - \frac{\dot{a}\,A'}{a\,A} = 0$$

$$R_{11}:\quad \frac{2\,c^2\,\dot{a}^2}{a^2\,A} + \frac{c^2\,\ddot{a}}{a\,A} - \frac{2\,\mathbb{C}'}{a^2\,r\,\mathbb{C}} + \frac{(\mathbb{C}')^2}{2\,a^2\,\mathbb{C}^2} + \frac{(A')^2}{4\,a^2\,A^2} + \frac{(\mathcal{D}_4')^2}{2\,a^2\,\mathcal{D}_4^2} - \frac{\mathbb{C}''}{a^2\,\mathbb{C}} - \frac{A''}{2\,a^2\,A} - \frac{\mathcal{D}_4''}{a^2\,\mathcal{D}_4} = \Lambda$$

$$R_{22}:\quad -\frac{1}{a^2\,r^2} + \frac{1}{r^2\,\mathbb{C}} - \frac{2\,\mathbb{C}'}{a^2\,r\,\mathbb{C}} - \frac{A'}{2\,a^2\,r\,A} - \frac{\mathbb{C}'\,A'}{4\,a^2\,\mathbb{C}\,A} - \frac{\mathcal{D}_4'}{a^2\,r\,\mathcal{D}_4} - \frac{\mathbb{C}'\,\mathcal{D}_4'}{2\,a^2\,\mathbb{C}\,\mathcal{D}_4} - \frac{\mathbb{C}''}{2\,a^2\,\mathbb{C}} = \Lambda$$

$$R_{44}:\quad \frac{2\,c^2\,\dot{a}^2}{a^2\,A} + \frac{c^2\,\ddot{a}}{a\,A} - \frac{\mathcal{D}_4'}{a^2\,r\,\mathcal{D}_4} - \frac{\mathbb{C}'\,\mathcal{D}_4'}{2\,a^2\,\mathbb{C}\,\mathcal{D}_4} - \frac{A'\,\mathcal{D}_4'}{4\,a^2\,A\,\mathcal{D}_4} - \frac{\mathcal{D}_4''}{2\,a^2\,\mathcal{D}_4} = \Lambda$$

We can simplify R_{01} by factoring out $\dfrac{\dot{a}}{a}$:

$$R_{01}:\quad \frac{2}{r} + \frac{\mathbb{C}'}{\mathbb{C}} + \frac{A'}{A} = 0$$

We can solve R_{01} either for A in terms of $\mathbb{C}$ or $\mathbb{C}$ in terms of A. We will do the latter:

$$\mathbb{C}[r] = \frac{\sqrt{3}\,c\,\sqrt{\kappa}}{r^2\,\Lambda\,A[r]} \quad \text{where } \kappa \text{ is a constant of integration and } \Lambda\neq 0.$$

This eliminates R_{01}, and the remaining equations become

$$R_{00}: \quad \frac{3\,c^2\,\ddot{a}}{a\,A} + \frac{3\,(A')^2}{4\,a^2\,A^2} - \frac{A'\,D_4'}{2\,a^2\,A\,D_4} - \frac{A''}{2\,a^2\,A} = \Lambda$$

$$R_{11}: \quad \frac{2\,c^2\,\dot{a}^2}{a^2\,A} + \frac{c^2\,\ddot{a}}{a\,A} - \frac{5\,(A')^2}{4\,a^2\,A^2} + \frac{(D_4')^2}{2\,a^2\,D_4^2} + \frac{A''}{2\,a^2\,A} - \frac{D_4''}{a^2\,D_4} = \Lambda$$

$$R_{22}: \quad \frac{A\,\Lambda}{\sqrt{3}\,c\,\sqrt{\kappa}} - \frac{3\,(A')^2}{4\,a^2\,A^2} + \frac{A'\,D_4'}{2\,a^2\,A\,D_4} + \frac{A''}{2\,a^2\,A} = \Lambda$$

$$R_{44}: \quad \frac{2\,c^2\,\dot{a}^2}{a^2\,A} + \frac{c^2\,\ddot{a}}{a\,A} + \frac{A'\,D_4'}{4\,a^2\,A\,D_4} - \frac{D_4''}{2\,a^2\,D_4} = \Lambda$$

We can algebraically solve $R_{00}+R_{22}$ for A:

$$R_{00}+R_{22}: \quad \frac{A\,\Lambda}{\sqrt{3}\,c\,\sqrt{\kappa}} + \frac{3\,c^2\,\ddot{a}}{a\,A} = 2\,\Lambda$$

$$\Rightarrow \quad A(r) = \frac{\sqrt{3}\left(c\,\sqrt{\kappa}\,\Lambda\,a[t] + \sqrt{c^2\,\sqrt{\kappa}\,\Lambda\,a[t]\left(\sqrt{\kappa}\,\Lambda\,a[t] - \sqrt{3}\,c\,\ddot{a}[t]\right)}\right)}{\Lambda\,a[t]}.$$

Observe that there are no r terms on the RHS,
so if the t-terms cancel out, that will make $A(r)$ a constant.

In order for the t-terms to cancel out, the $a(t)$
terms in the numerator must cancel with the one $a(t)$ term in the
denominator. That occurs precisely when $\ddot{a}(t)$ is proportional to $a(t)$.

Solving $\ddot{a}(t) = \zeta\,a(t)$, where ζ is a proportionality constant, yields

$$a(t) = e^{t\sqrt{\zeta}}\,\alpha + e^{-t\sqrt{\zeta}}\,\beta$$

where α and β are constants of integration, not both zero.

With this value of $a(t)$, the $a(t)$ terms in the numerator
cancel with the $a(t)$ term in the denominator, leaving $A(r)$ a constant:

$A(r) = K$, a constant, K.

With this value for $A(r)$, equations R_{00} and
R_{22} are dependent, and we are free to declare R_{00} redundant.
This reduces our situation to 3 equations in the one unknown, D_4:

$$R_{11}: \quad \frac{2\,c^2\,\dot{a}^2}{a^2\,K} + \frac{c^2\,\ddot{a}}{a\,K} + \frac{(D_4')^2}{2\,a^2\,D_4^2} - \frac{D_4''}{a^2\,D_4} = \Lambda$$

$$R_{22}: \quad \frac{K\,\Lambda}{\sqrt{3}\,c\,\sqrt{\kappa}} = \Lambda$$

$$R_{44}: \quad \frac{2\,c^2\,\dot{a}^2}{a^2\,K} + \frac{c^2\,\ddot{a}}{a\,K} - \frac{D_4''}{2\,a^2\,D_4} = \Lambda$$

No mattter whether we solve for D_4 from R_{11}, R_{44}, or $R_{11}-2R_{44}$, when we plug in $a(t)$ it is not obvious that $D_4(r)$ is strictly a function of r. So, we solve D_4 from $R_{11}-2R_{44}$ and defer checking whether D_4 is strictly a function of r:

$$R_{11}-2R_{44}: \quad -\frac{2\,c^2\,\dot{a}^2}{a^2\,K} - \frac{c^2\,\ddot{a}}{a\,K} + \frac{(D_4')^2}{2\,a^2\,D_4^2} = -\Lambda$$

$$D_4(r) = e^{\pm\frac{i\,\sqrt{2}\,r\,\sqrt{a^2\,K\,\Lambda-2\,c^2\,\dot{a}^2-a\,c^2\,\ddot{a}}}{\sqrt{K}}}\,k \quad \text{where k is a constant of integration.}$$

For now, assume D_4 is strictly a function of r. Then there is a constant $\mathcal{L}$ such that

$$D_4(r) = e^{\mathcal{L}\,r}$$

Since $R_{11}-2R_{44}=-\Lambda$, these equations are dependent and we are free to designate R_{44} to be redundant. Then we are left with two equations:

$$R_{11}: \quad -\frac{\mathcal{L}^2}{2\,a[t]^2} + \frac{2\,c^2\,a'[t]^2}{K\,a[t]^2} + \frac{c^2\,a''[t]}{K\,a[t]} = \Lambda$$

$$R_{22}: \quad \frac{K\,\Lambda}{\sqrt{3}\,c\,\sqrt{K}} = \Lambda$$

When we plug in $a(t)$, we get

$$R_{11}: \quad -\frac{e^{2\,t\,\sqrt{\zeta}}\,\mathcal{L}^2}{2\left(e^{2\,t\,\sqrt{\zeta}}\,\alpha+\beta\right)^2} + \frac{3\,c^2\,e^{4\,t\,\sqrt{\zeta}}\,\alpha^2\,\zeta}{\left(e^{2\,t\,\sqrt{\zeta}}\,\alpha+\beta\right)^2 K} - \frac{2\,c^2\,e^{2\,t\,\sqrt{\zeta}}\,\alpha\,\beta\,\zeta}{\left(e^{2\,t\,\sqrt{\zeta}}\,\alpha+\beta\right)^2 K} + \frac{3\,c^2\,\beta^2\,\zeta}{\left(e^{2\,t\,\sqrt{\zeta}}\,\alpha+\beta\right)^2 K} = \Lambda$$

I am unable to definitively find all the solutions to this for every combination of constants, but it seems likely that the only way to eliminate t is to set

$$\zeta = 0.$$

$$\Rightarrow \quad R_{11}: \quad -\frac{\mathcal{L}^2}{2\,(\alpha+\beta)^2} = \Lambda$$

$$\Rightarrow \quad \mathcal{L} = \pm i\,\sqrt{2}\,(\alpha+\beta)\,\sqrt{\Lambda}$$

R_{11} is now satisfied.

Moreover, $a(t)$ has become a constant:

$$a(t) = \alpha+\beta.$$

$$\Rightarrow \quad D_4(r) = e^{i\,\sqrt{2}\,r\,(\alpha+\beta)\,\sqrt{\Lambda}}.$$

We have confirmed the assumption that $D_4(r)$ is strictly a function of r. ✓

We can solve R_{22}, the last remaining equation, for κ:

$$R_{22}: \quad \frac{K\Lambda}{\sqrt{3}\, c\, \sqrt{\kappa}} = \Lambda$$

$$\kappa = \frac{K^2}{3\, c^2}$$

All of the equations are now satisfied.

Since a(t) now equals the constant, $\alpha + \beta$, we can further simplify the equations by setting a(t)=α

This solution to the Λ field equations is

$a(t) = \alpha$ where α is a free constant,

$A(t,r) = K$ where K is a free constant

$B(t,r) = a[t]^2$,

$\mathbb{C}(t,r) = \frac{1}{r^2 \Lambda}$ where $\Lambda \neq 0$,

$\mathbb{D}_4(r) = e^{\pm(i \sqrt{2}\, r\, \alpha\, \sqrt{\Lambda})}$

This solution is new, neither a special case nor a generalization of the Case d solution:

- In the Case d solution, the free parameter a(t) cannot be a constant or else A(t,r)=0, making $g_{00}=0$, not a metric.
- So, Case d_Iα is a new solution even though $\mathbb{B}$ and $\mathbb{C}$ are the same and the $\mathbb{D}_4$'s are similar exponential functions.

d_II: B(t,r)=a(t)^2, $\mathbb{C}$(t,r)=$\mathbb{C}$(r), A(t,r)=c^2

One solution to this case, which we label 'Solution 1', comes from plugging A(t,r)=c^2 into the solution to Case d:

SOLUTION 1

$a(t) = \pm \frac{\sqrt{3}\, c\, \sqrt{\kappa}\, \mathrm{Sinh}\left[\frac{\sqrt{\Lambda}\,(t-\upsilon)}{\sqrt{3}}\right]}{\Lambda}$ where κ and υ are free constants,

$A(t,r) = c^2$,

$B(t,r) = a(t)^2$,

$\mathbb{C}(t,r) = \frac{1}{r^2 \Lambda}$ where $\Lambda \neq 0$,

$\mathbb{D}_4(r) = e^{\pm \frac{2\, c\, r\, \sqrt{\kappa}}{\sqrt{\Lambda}}}$

This solution, of course, is a special case of the Case d solution. However, it is neither a special case nor a generalization of Case d_Iα.

It is conceivable that we might get a different solution if we set $A(t,r) = c^2$ before we solve the ODEs rather than afterwards. Hence, we work this case from scratch to find out.

Search for other solutions:

Starting fresh, the field equations become

$$R_{00}: \quad \frac{3\,\ddot{a}}{a} = \Lambda$$

$$R_{01}: \quad -\frac{2\,\dot{a}}{a\,r} - \frac{\dot{a}\,C'}{a\,C} = 0$$

$$R_{11}: \quad \frac{2\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} - \frac{2\,C'}{a^2\,r\,C} + \frac{(C')^2}{2\,a^2\,C^2} + \frac{(D_4')^2}{2\,a^2\,D_4^2} - \frac{C''}{a^2\,C} - \frac{D_4''}{a^2\,D_4} = \Lambda$$

$$R_{22}: \quad -\frac{1}{a^2\,r^2} + \frac{1}{r^2\,C} - \frac{2\,C'}{a^2\,r\,C} - \frac{D_4'}{a^2\,r\,D_4} - \frac{C'\,D_4'}{2\,a^2\,C\,D_4} - \frac{C''}{2\,a^2\,C} = \Lambda$$

$$R_{44}: \quad \frac{2\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} - \frac{D_4'}{a^2\,r\,D_4} - \frac{C'\,D_4'}{2\,a^2\,C\,D_4} - \frac{D_4''}{2\,a^2\,D_4} = \Lambda$$

We can simplify R_{01} by factoring out $\frac{\dot{a}}{a}$:

$$R_{01}: \quad \frac{2}{r} + \frac{C'}{C} = 0$$

Solving for C yields:

$$C[r] = \frac{K}{r^2}$$ where K is a constant of integration.

This eliminates R_{01}, and the remaining equations become

$$R_{00}: \quad \frac{3\,\ddot{a}}{a} = \Lambda$$

$$R_{11}: \quad \frac{2\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} + \frac{(D_4')^2}{2\,a^2\,D_4^2} - \frac{D_4''}{a^2\,D_4} = \Lambda$$

$$R_{22}: \quad \frac{1}{K} = \Lambda$$

$$R_{44}: \quad \frac{2\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} - \frac{D_4''}{2\,a^2\,D_4} = \Lambda$$

From R_{22} we get

$$\Lambda \neq 0$$

and

$$K = \frac{1}{\Lambda}.$$

Moreover, R_{00} shows that a(t) is not generic for this solution:

$$a(t) = e^{\frac{t\sqrt{\Lambda}}{\sqrt{3}}}\alpha + e^{-\frac{t\sqrt{\Lambda}}{\sqrt{3}}}\beta$$ where α and β are constants of integration.

R_{00} and R_{22} have been solved, leaving only R_{11} and R_{44}.

For the sake of keeping the equations more manageable,
we work with the symbol 'a(t)' for now rather than its formula.

$R_{11}-2R_{44}$ can be solved for D_4:

$$R_{11}-2R_{44}: \quad -\frac{2\,\dot{a}^2}{a^2} - \frac{\ddot{a}}{a} + \frac{(D_4')^2}{2\,a^2\,D_4^2} = -\Lambda$$

$$D_4(r) = e^{\pm\left(i\sqrt{2}\,r\sqrt{\Lambda\,a[t]^2-2\,\dot{a}[t]^2-a[t]\,\ddot{a}[t]}\right)}\xi$$ where ξ is a constant of integration.

As in Case d, we can roll ξ into the radius p_4:

$$D_4(r) = e^{\pm\left(i\sqrt{2}\,r\sqrt{\Lambda\,a[t]^2-2\,\dot{a}[t]^2-a[t]\,\ddot{a}[t]}\right)}$$

Plugging in the formula for a(t), we
confirm that this really is a function strictly of r:

$$D_4(r) = e^{\pm\frac{4\,i\,r\sqrt{\alpha\,\beta\,\Lambda}}{\sqrt{3}}}$$

Plugging this in, we see that the remaining two equations are satisfied:

R_{11}: $\Lambda = \Lambda$

R_{44}: $\Lambda = \Lambda$

All of the equations are satisfied,
so we have an exact solution to the $\Lambda\neq 0$ field equations:

SOLUTION 2

$$a(t) = e^{\frac{t\sqrt{\Lambda}}{\sqrt{3}}}\alpha + e^{-\frac{t\sqrt{\Lambda}}{\sqrt{3}}}\beta$$ where α and β are free constants,

$$A(t,r) = c^2,$$

$$B(t,r) = a[t]^2,$$

$$\mathbb{C}(t,r) = \frac{1}{r^2\,\Lambda} \quad \text{where } \Lambda\neq 0,$$

$$\mathbb{D}_4(t,r) = \mathbb{D}_5(t,r) = a(t)^2\mathcal{D}_4(r) = a[t]^2 e^{\pm\frac{4\,i\,r\,\sqrt{\alpha\,\beta\,\Lambda}}{\sqrt{3}}},$$

and line element

$$ds^2 = c^2\,dt^2 - a[t]^2\,dr^2 - \frac{1}{\Lambda}\,d\theta^2 - \frac{\mathrm{Sin}[\theta]^2}{\Lambda}\,d\phi^2 - e^{\pm\frac{4\,i\,r\,\sqrt{\alpha\,\beta\,\Lambda}}{\sqrt{3}}}\,a[t]^2\,p_4^2\,d\psi_4^2 - e^{\pm\frac{4\,i\,r\,\sqrt{\alpha\,\beta\,\Lambda}}{\sqrt{3}}}\,a[t]^2\,p_5^2\,d\psi_5^2$$

This solution is new. Claim it is neither a special case nor a generalization of either the Case d or Case d_Iα solutions:

First, compare it to Case d:

- Since $a(t)$ in Case d is a free parameter, $a(t)$ in Solution 2 is a special case of it.
- In order for $A(t,r)=c^2$ in Solution 2 to be a special case of Case d, Solution 2 must be a special case of of Solution 1 because Solution 1 is the special case of Case d having $A(t,r)=c^2$.
- But, $a(t)=e^{\frac{t\sqrt{\Lambda}}{\sqrt{3}}}\alpha + e^{-\frac{t\sqrt{\Lambda}}{\sqrt{3}}}\beta$ of Solution 2 is not a special case of $a(t)=\dfrac{\sqrt{3}\,c\,\sqrt{\kappa}\,\mathrm{Sinh}\left[\frac{\sqrt{\Lambda}\,(t-\upsilon)}{\sqrt{3}}\right]}{\Lambda}$ of Solution 1:
  - ✓ In Solution 2,

    $$a(t) = e^{\frac{t\sqrt{\Lambda}}{\sqrt{3}}}\alpha + e^{-\frac{t\sqrt{\Lambda}}{\sqrt{3}}}\beta\ ]$$
    $$= \alpha\mathrm{Cosh}[\frac{t\sqrt{\Lambda}}{\sqrt{3}}] + \beta\mathrm{Cosh}[\frac{t\sqrt{\Lambda}}{\sqrt{3}}] + \alpha\mathrm{Sinh}[\frac{t\sqrt{\Lambda}}{\sqrt{3}}] - \beta\mathrm{Sinh}[\frac{t\sqrt{\Lambda}}{\sqrt{3}}]$$
  - ✓ υ in Solution 1 must be zero so that $a(t)=\sqrt{3\,\kappa}\,\frac{c}{\Lambda}\mathrm{Sinh}[\frac{\sqrt{\Lambda}\,t}{\sqrt{3}}]$

    That leaves Solution 1 only 1 free constant, κ.
  - ✓ But, Solution 2 has two free constants, α and β.
  - ✓ So, Solution 2 cannot be a special case of Solution 1.
  - ✓ Hence, Solution 2 is not a special case of Case d.
- Solution 2 also is not a generalization of Solution 1 because of constant υ in $a(t)$ of Solution 1.

As for Case d_Iα, a(t) is constant. In order for the Solution 2 a(t) to be constant, we would have to have Λ=0 ⇒ $\mathbb{C}$(t,r) is undefined. So, Case d_Iα is not a special case of Solution 2. And, clearly the Case d_Iα constant a(t) does not generalize a(t) of Solution 2. So, neither solution is a special case of the other.

In viewing all four solutions together (Cases d and d_Iα and Solutions 1 and 2), we observe they are similar. All have the same B and $\mathbb{C}$, and essentially the same $\mathbb{D}_4$. They only differ in a(t) and/or A(t,r).